

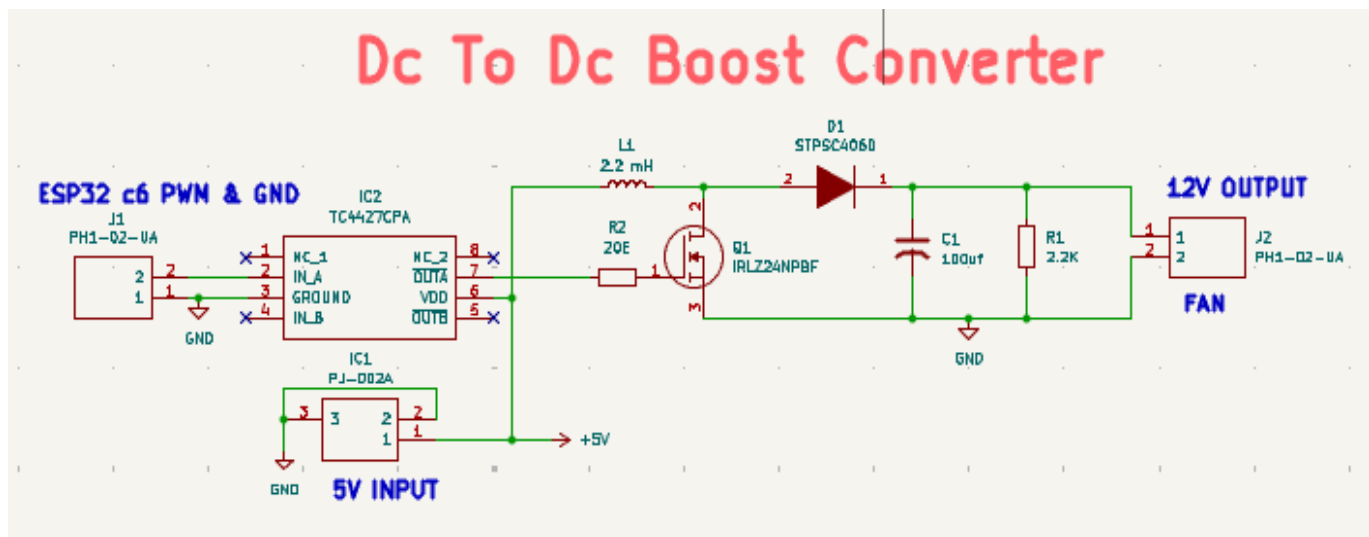
PCB Design for Dc - Dc Boost Converter

Outline

A DC–DC Boost Converter is a power electronic circuit that steps up (increases) a lower DC input voltage to a higher DC output voltage. It works using a switching device (like a MOSFET), inductor, diode, and capacitor.

The Dc-Dc Boost converter used here takes 5V input Voltage and steps up to 12V output.

Circuit diagram:

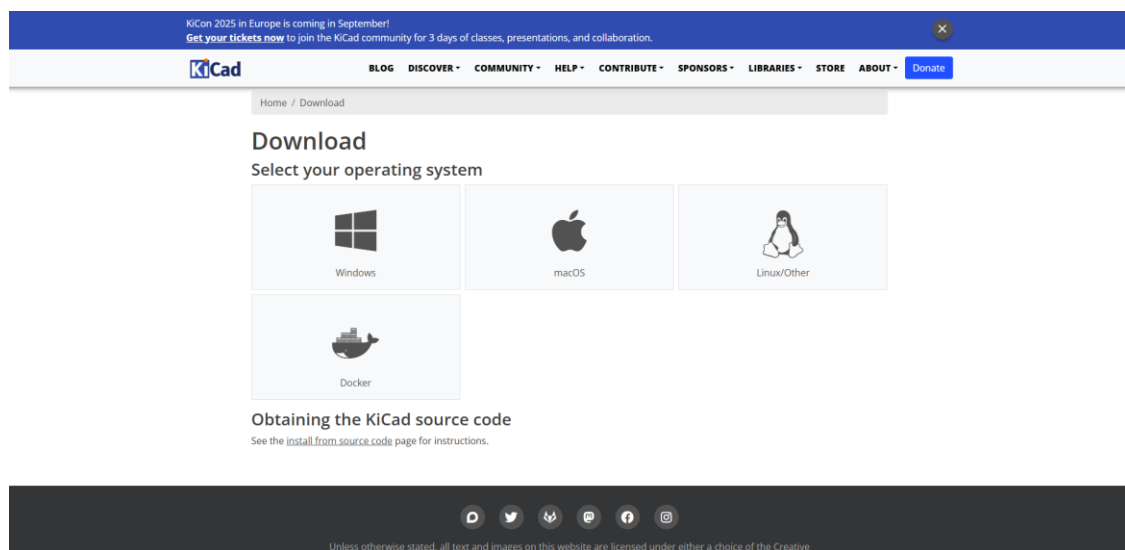


This Manual is a practical guide with step-by-step instructions from start to production ready Gerbers using KiCad. Intended for beginners and intermediate users who want a clean, repeatable PCB design workflow.-by-step user manual that walks you from a blank project to production-ready Gerbers using KiCad. Intended for beginners and intermediate users who want a clean, repeatable PCB design workflow.

I: General Instructions:

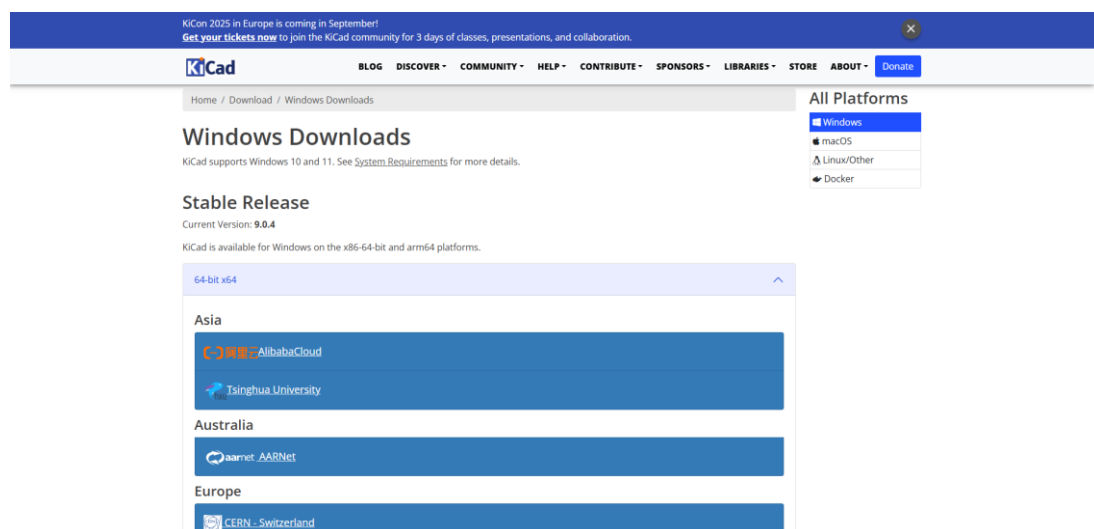
Download Kicad Software

Step 1: Go to the official KiCad website-[Link](#)



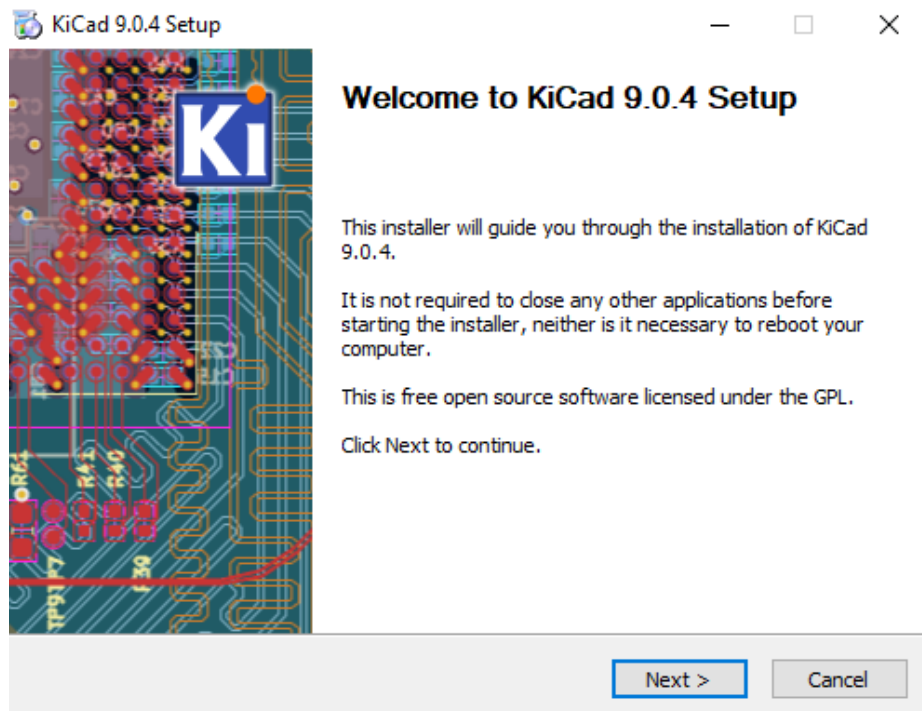
Step 2: Choose your Operating System

- ‘Windows’ → Download the installer (.exe)
- To select the Asia version.



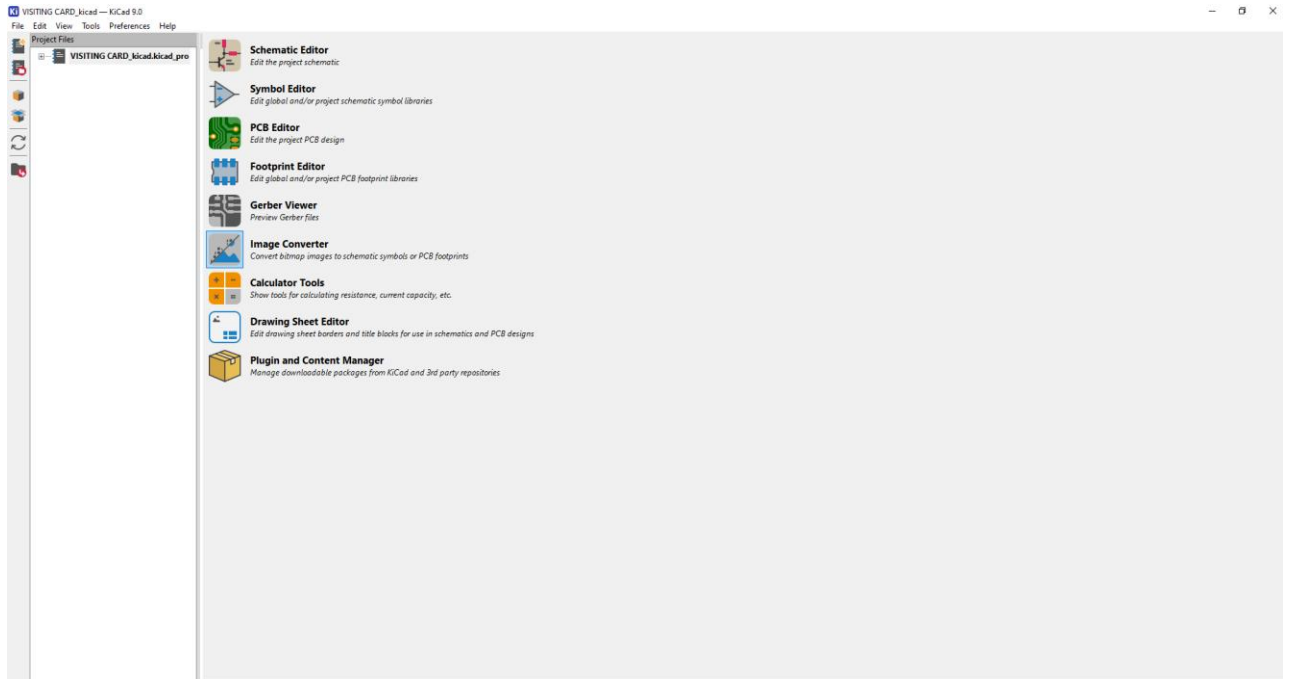
Step 3: Install

- Run the downloaded installer and follow the setup wizard.



Step 4: Verify installation

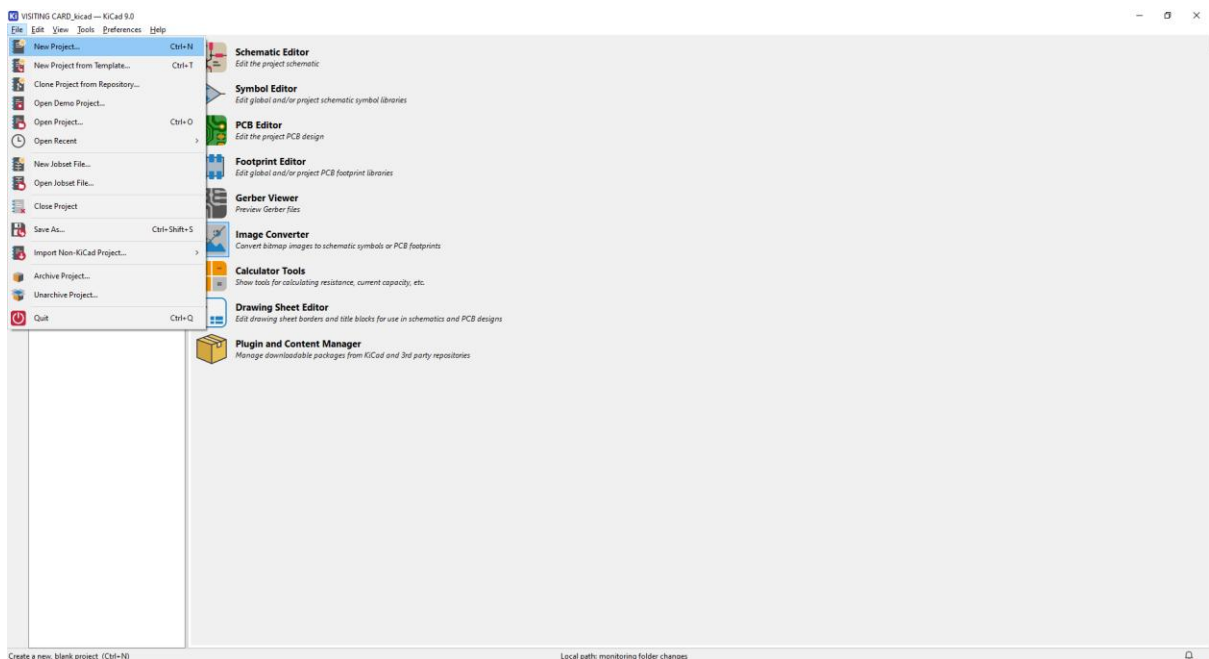
- Open KiCad from your applications menu.
- The KiCad Project Manager window should appear, showing buttons for Schematic Editor and PCB Editor.



Start a project:

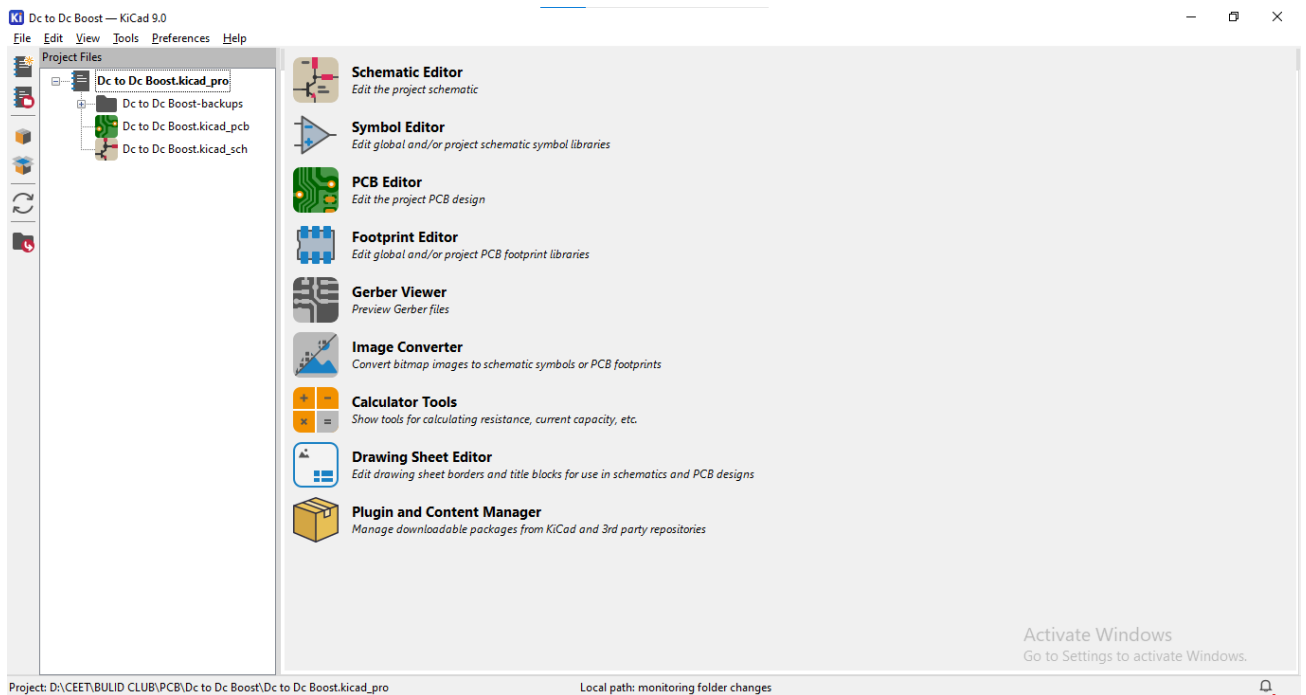
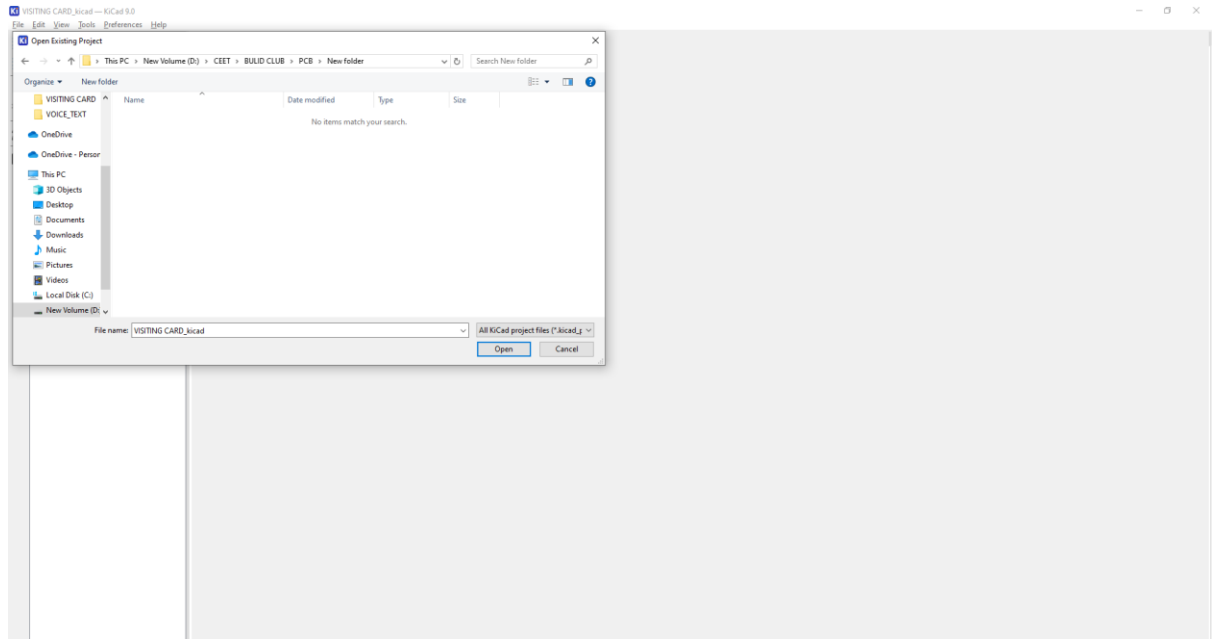
Step1:

- File → New Project → give a clear name



Step2:

- Enter the project title
- Save in a folder structure.



Step 3:

- Circuit components :

S. No	Components	Quantity
1.	2.2 mH Inductor	1
2.	100-470uf (63V) Capacitor	1
3.	22 ohms,2.2K ohm (Through hole) Resistor	2
4.	IRLZ24NPBF Mosfet	1
5.	STPSC406D Schottky Diode	1
6.	TC4427 Gate Drivers	1
7.	PJ-002A Power Connector	1
8.	PH1-02-UA Berg strip male connector	2

- Some components are not there in the kicad Ide so externally download the components after import the kicad ide.
- Create a mouser Account -[link](#)





[EXAMPLES](#)
[LOGIN](#)
[REGISTER](#)

Account Registration

Supplyframe/SamacSys is a third-party service provider to Mouser and a data controller that is collecting customer personal data for the purposes of updating you about component updates and changes to CAD models.
Please follow the link to the Supplyframe/Samacsys [Privacy Policy](#) for additional information.

Title

First Name

Last Name

Company

Country

State

Town

Job Role

Email

Password

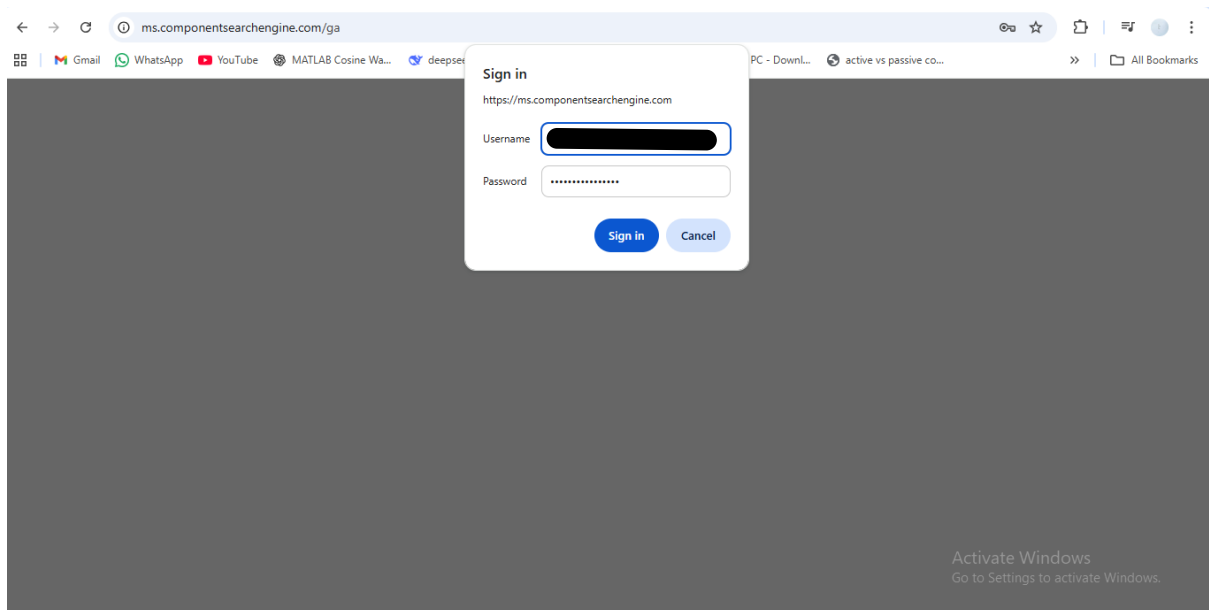
Confirm Password

Public Alias

I agree to the [Terms](#) and [Privacy Policy](#) ☐

Activate Windows
Go to Settings to activate Windows.

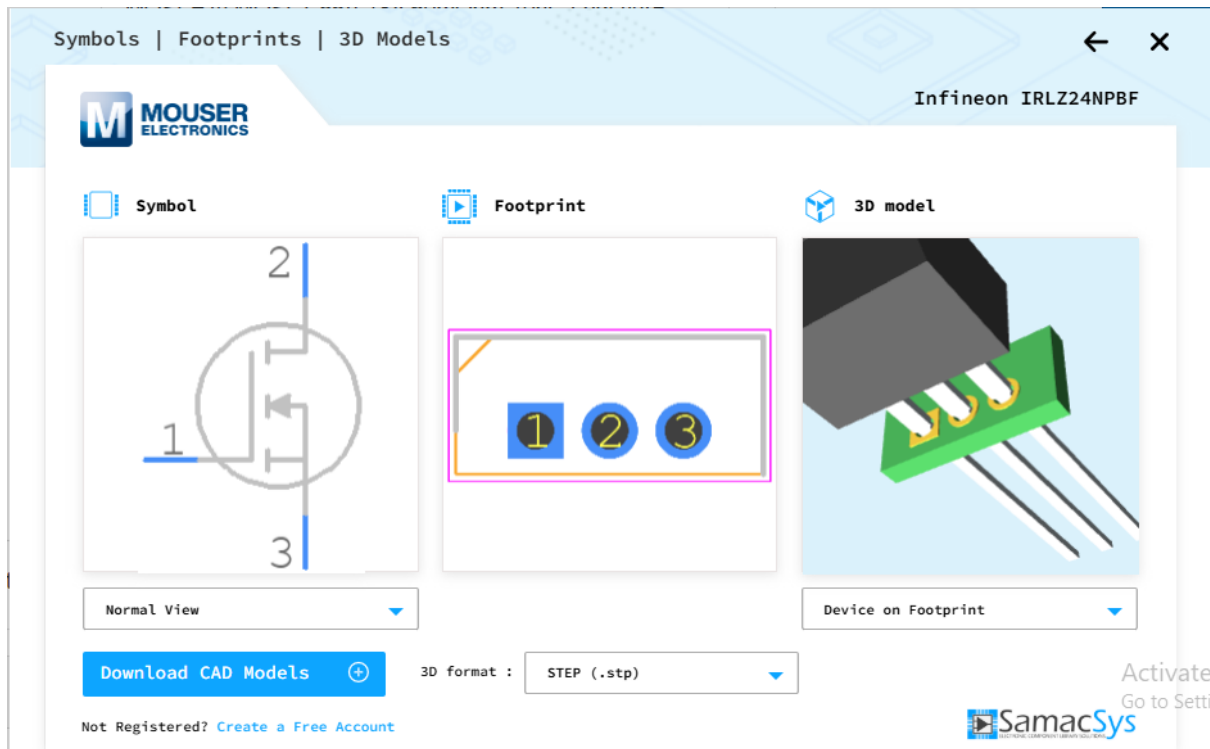
- Account Registration ones completed after going to login it



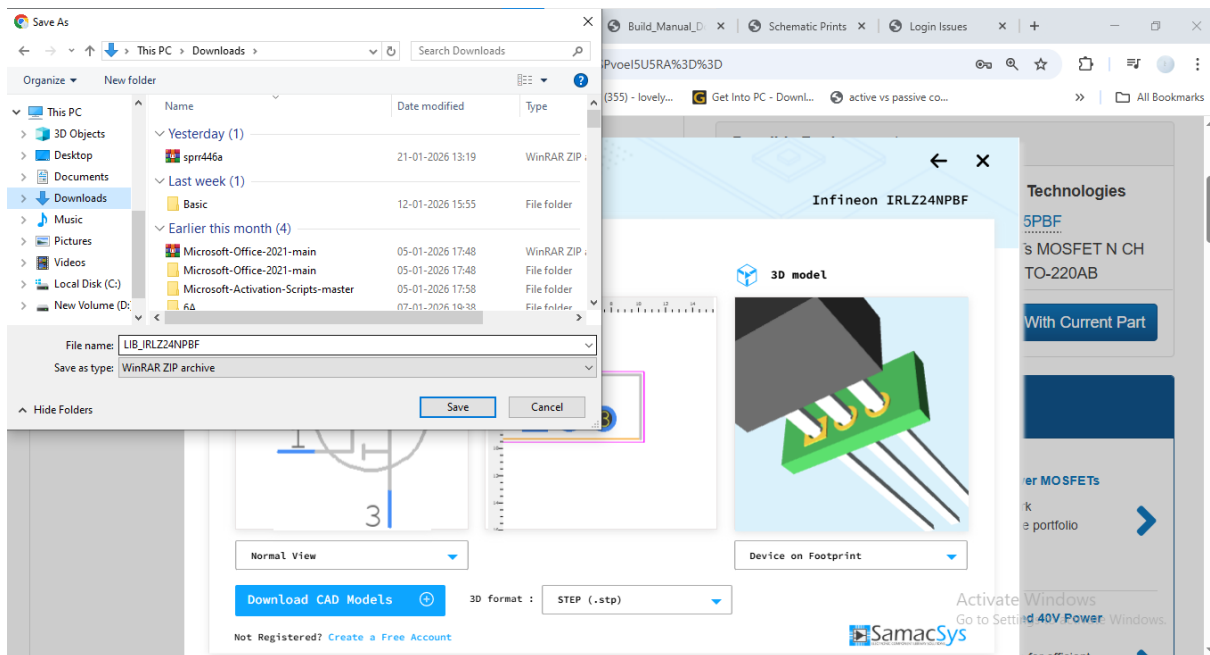
- IRLZ24NPBF Mosfet download- [link](#)

 Images are for reference only See Product Specifications Share	Mouser No: 942-IRLZ24NPBF	Stock: Restricted Availability
	Mfr. No: IRLZ24NPBF	Possible Replacement
	Mfr.: Infineon Technologies	
	Customer No: Customer No	 Mfr.: Infineon Technologies Mfr. No: IRFB7545PBF Description: MOSFETs MOSFET N CH 60V 95A TO-220AB Compare With Current Part
	Description: MOSFETs MOSFT 55V 18A 60mOhm 10nC LogLvIAB	FEATURED PRODUCTS INFINEON
	Lifecycle:  Obsolete	
Datasheet: IRLZ24NPBF Datasheet (PDF)	 OptiMOS™ 6 80V Power MOSFETs Sets industry benchmark performance with a wide portfolio offering. Learn More	
ECAD Model:  PCB Symbol, Footprint & 3D Model Download the free Library Loader to convert this file for		

- Click the ECAD Model option



- Click the Download CAD Model option

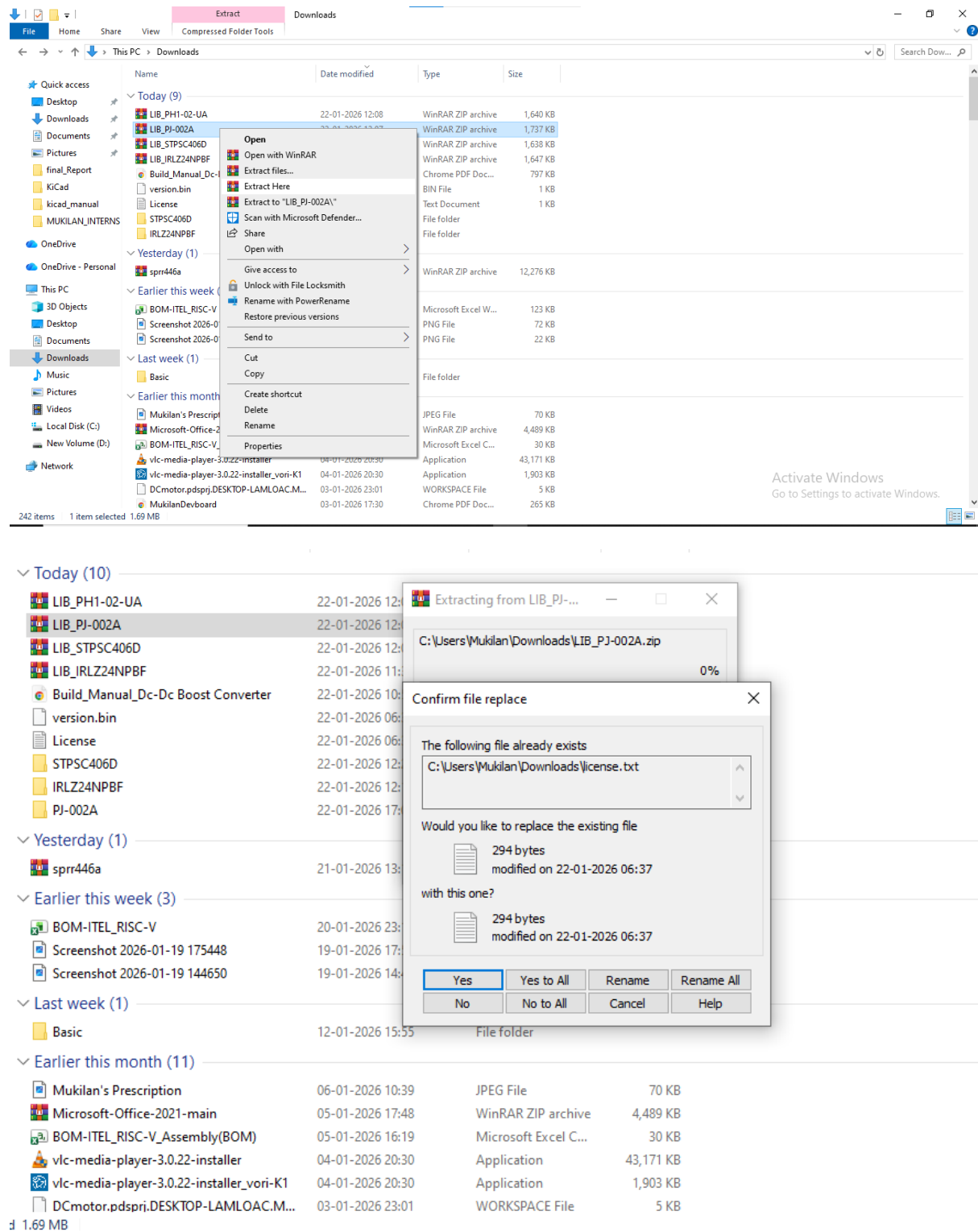


- Save the Zip file.
- IRLZ24NPBF Mosfet download- [link](#)
- STPSC406D Schottky Diode download- [link](#)
- TC4427 Gate Drivers download- [link](#)
- PJ-002A Power Connector download- [link](#)
- PH1-02-UA Berg strip male connector download- [link](#)

After following all the steps, the components were download it

Step 4:

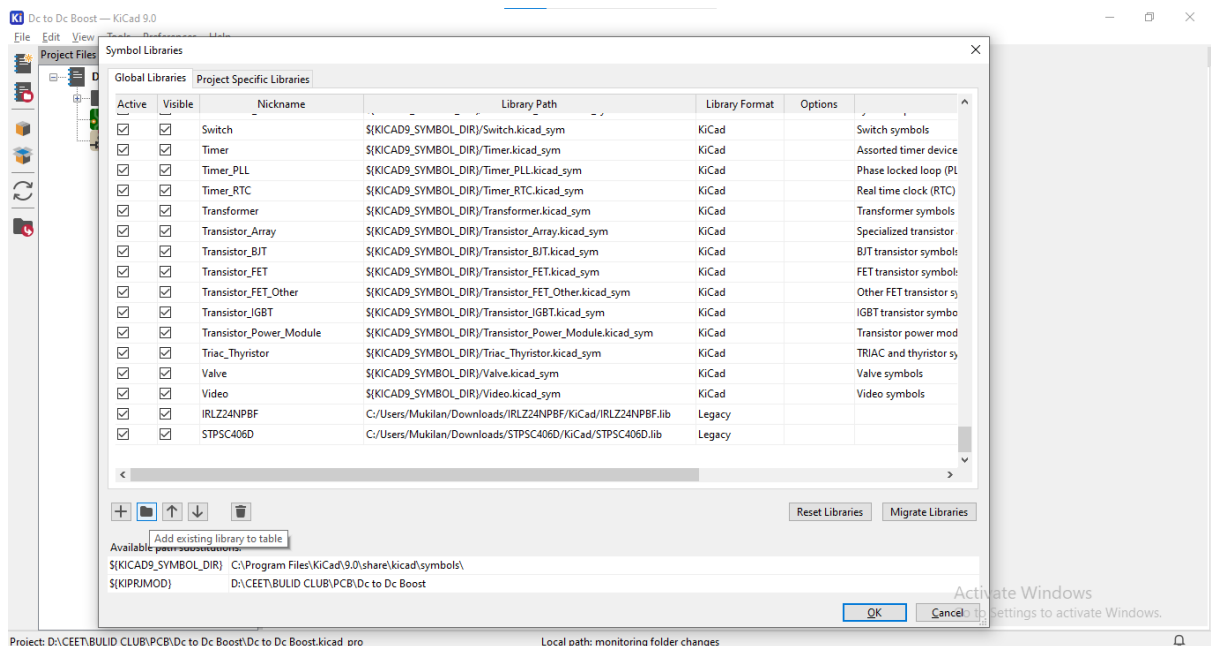
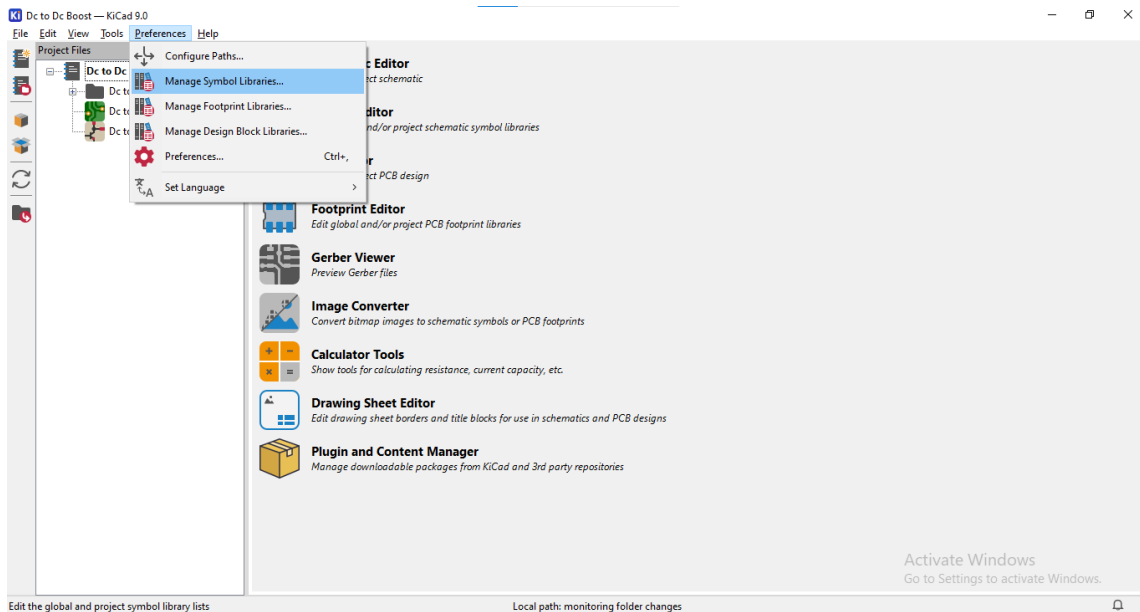
- Download the components after extracting the ZIP file.



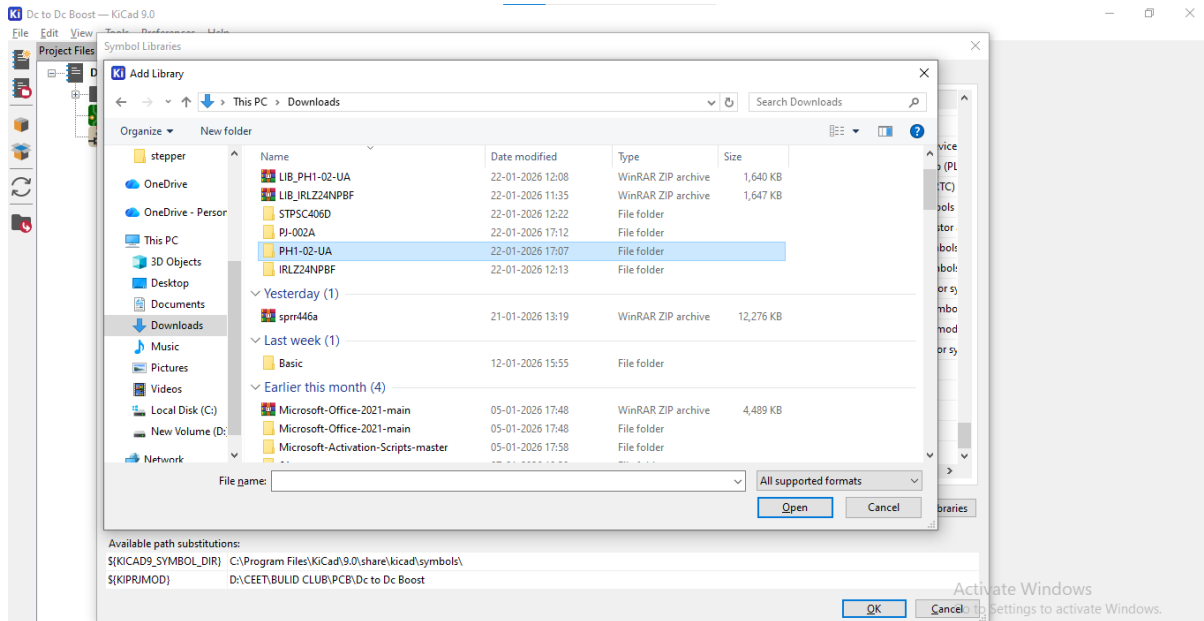
- To continue yes to all click it
- Please follow all the steps and extract the ZIP file

Step 5:

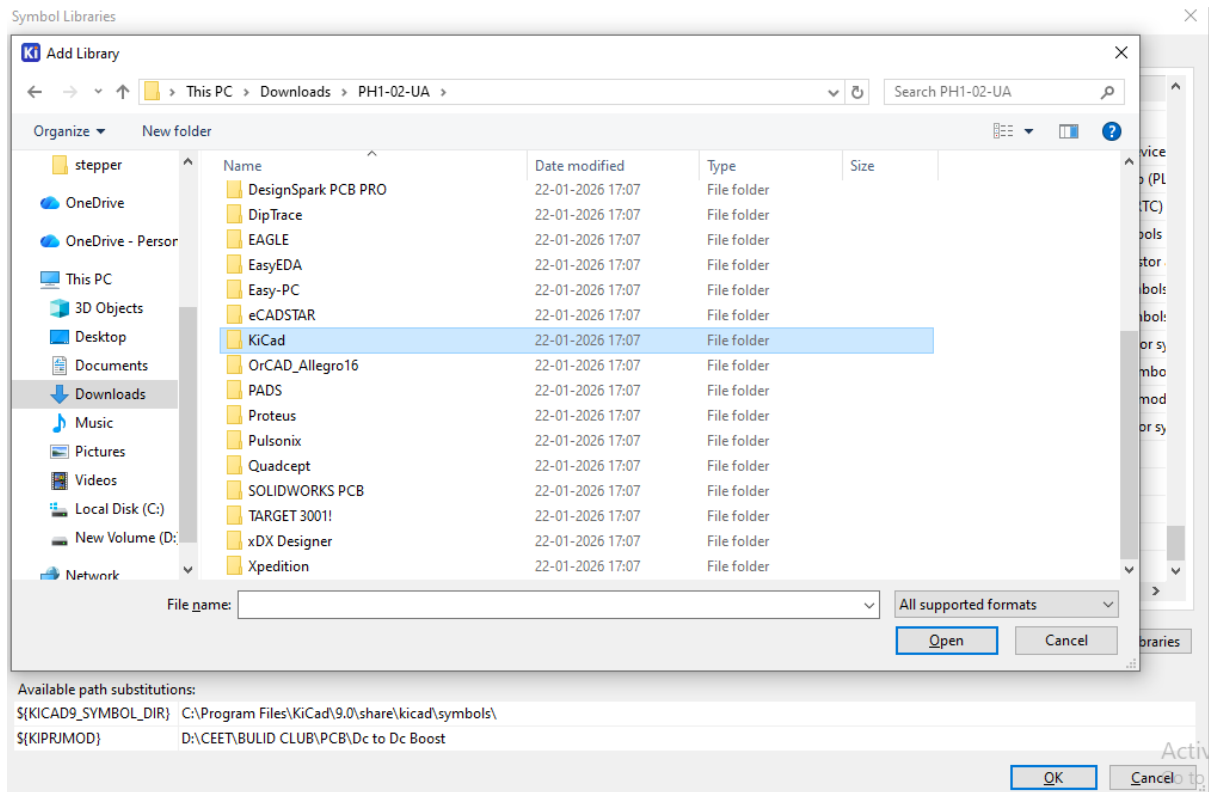
- Go to Preferences > Manage Symbol Libraries Open it



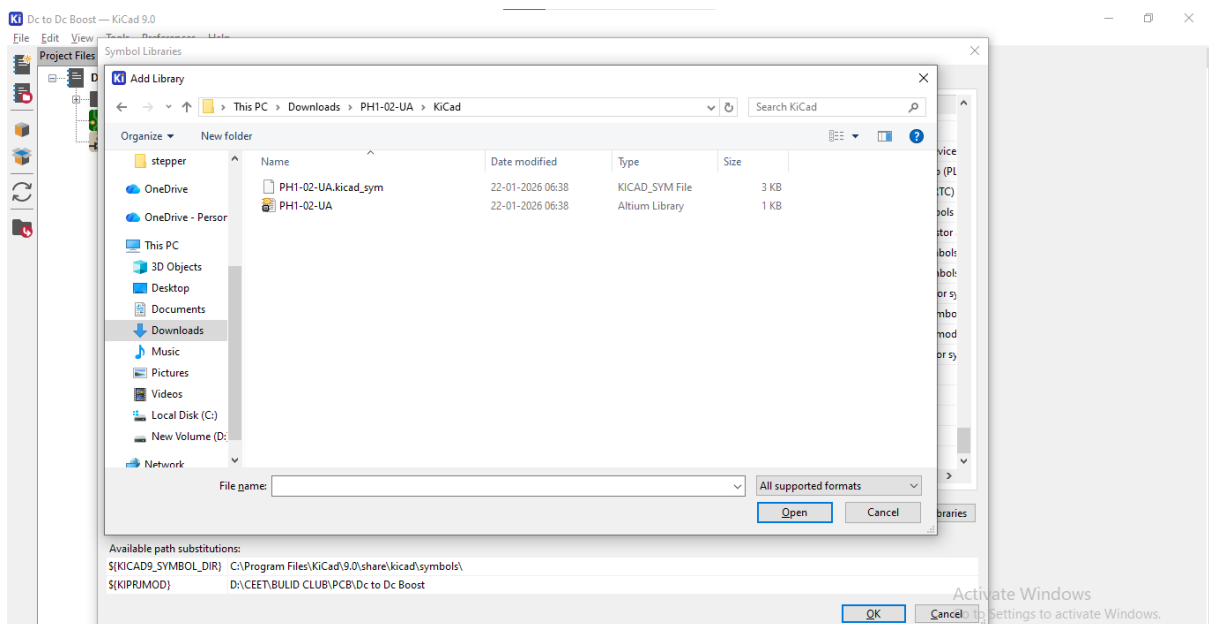
- Add existing library to table option has been open it.



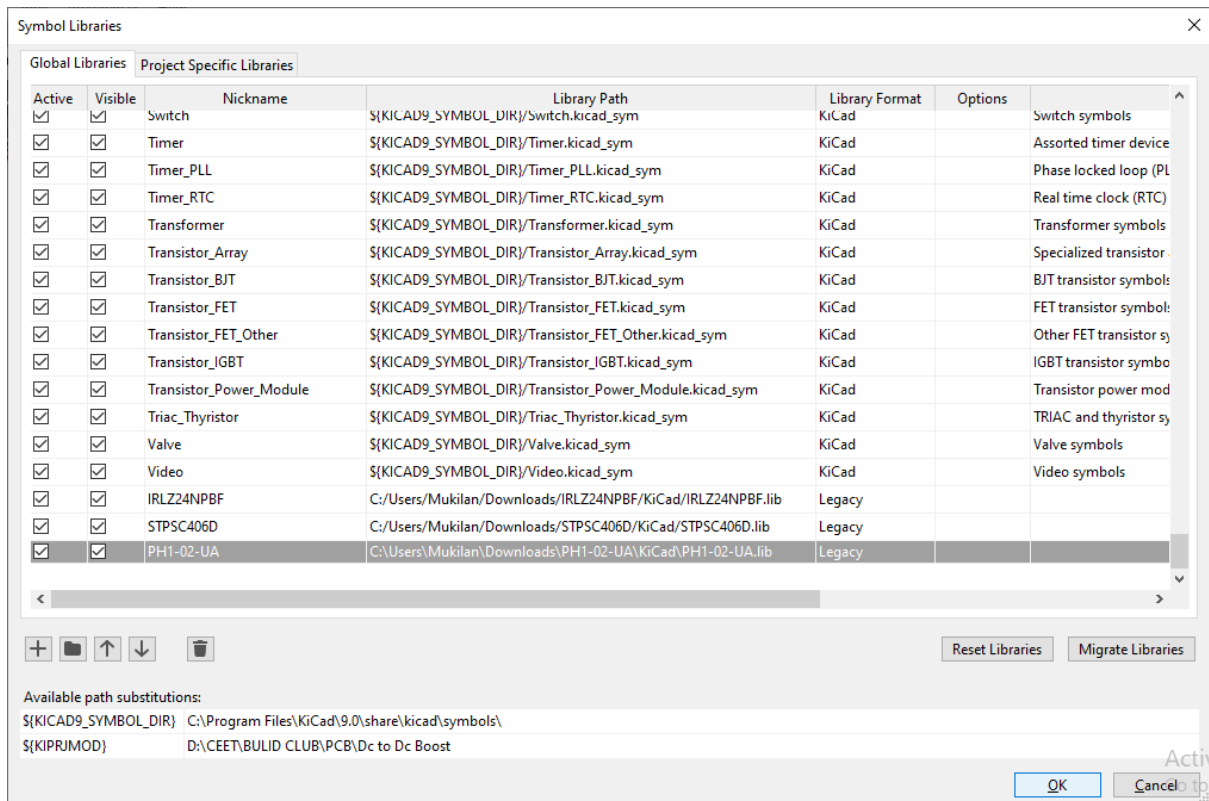
- Open the components folder, select the kicad folder.



- Select the Kicad Library file



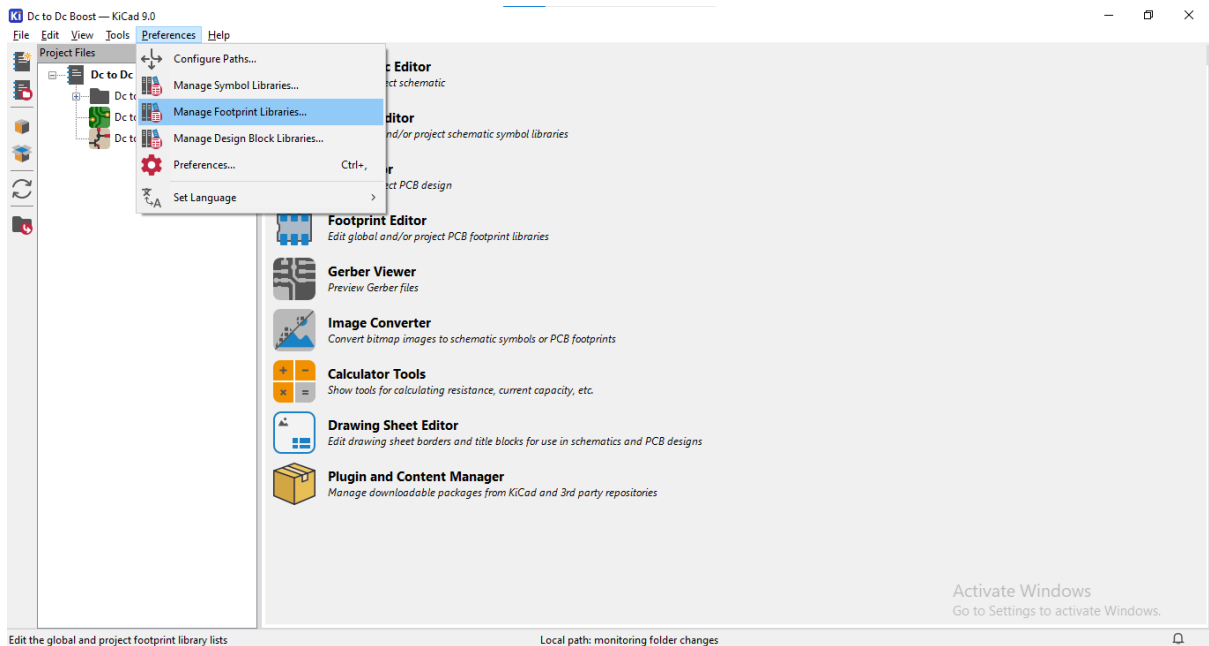
- After Symbol Libraries window has ok option click it.

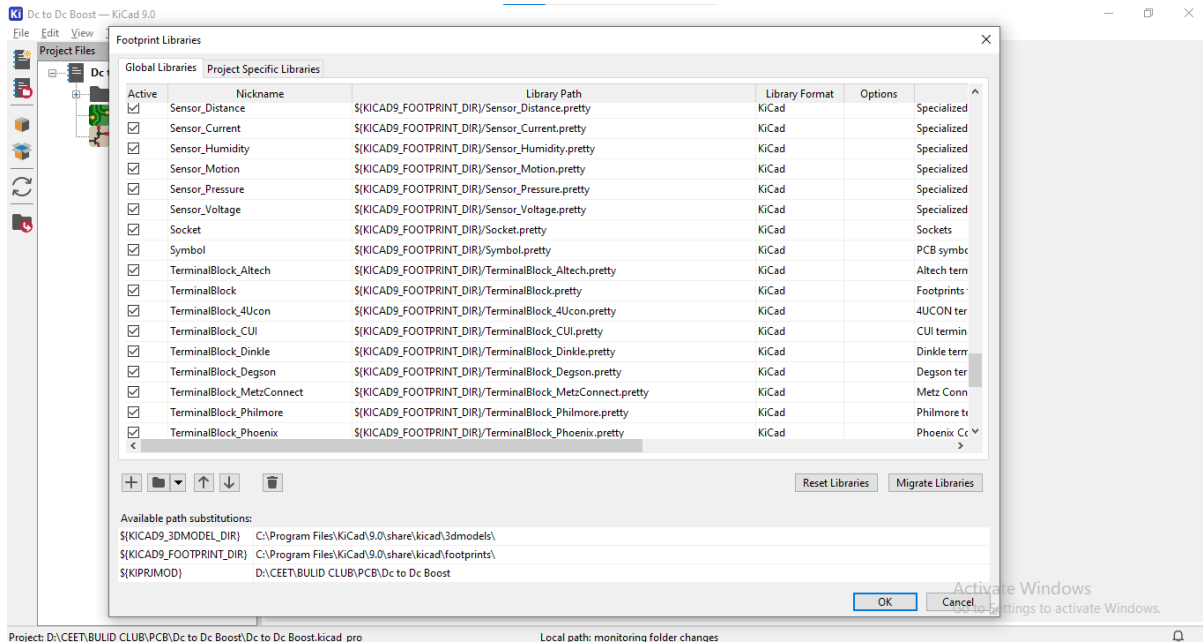


Please follow all the steps to import all components

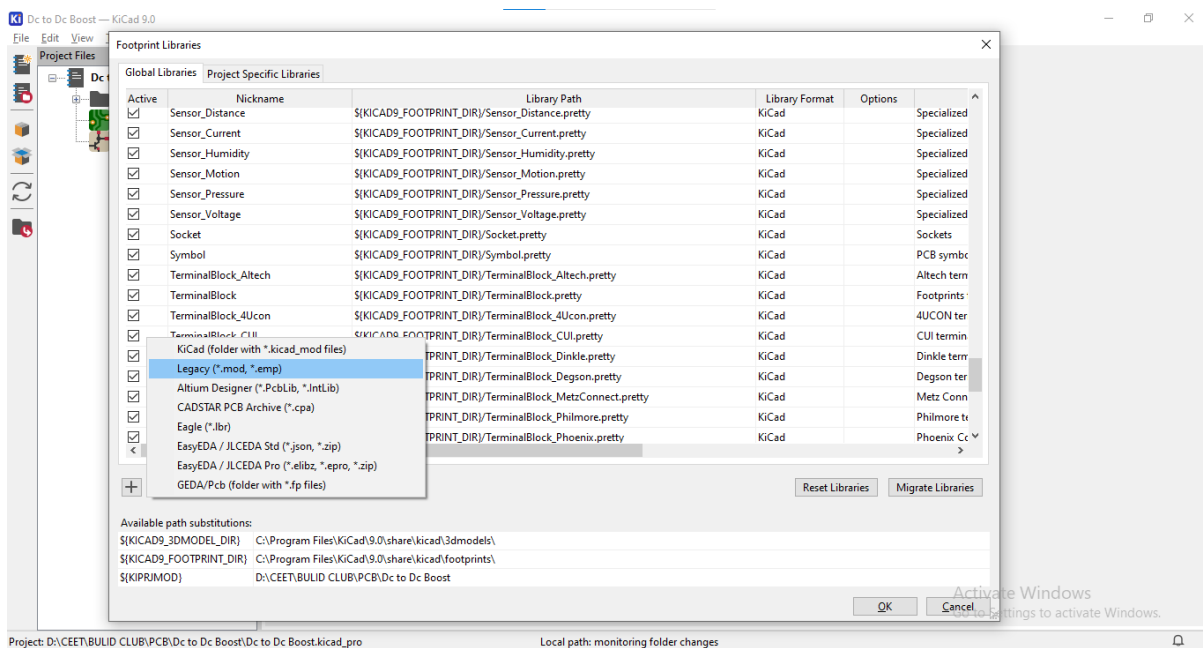
Step 6:

- Go to Preferences > Manage Footprint Libraries Open it

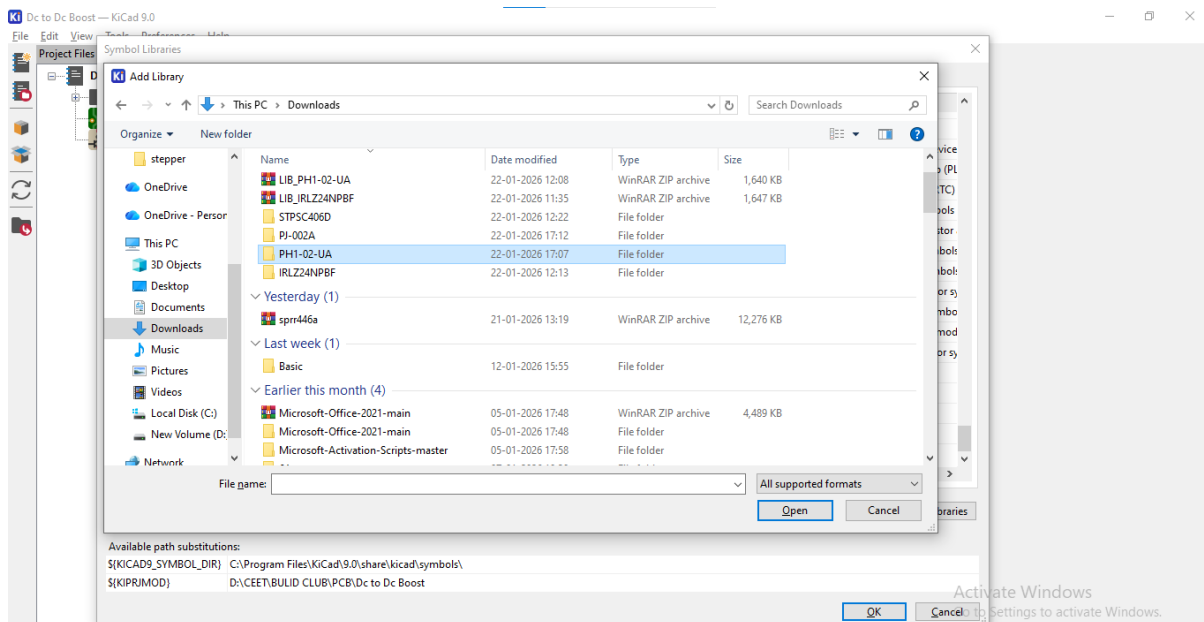




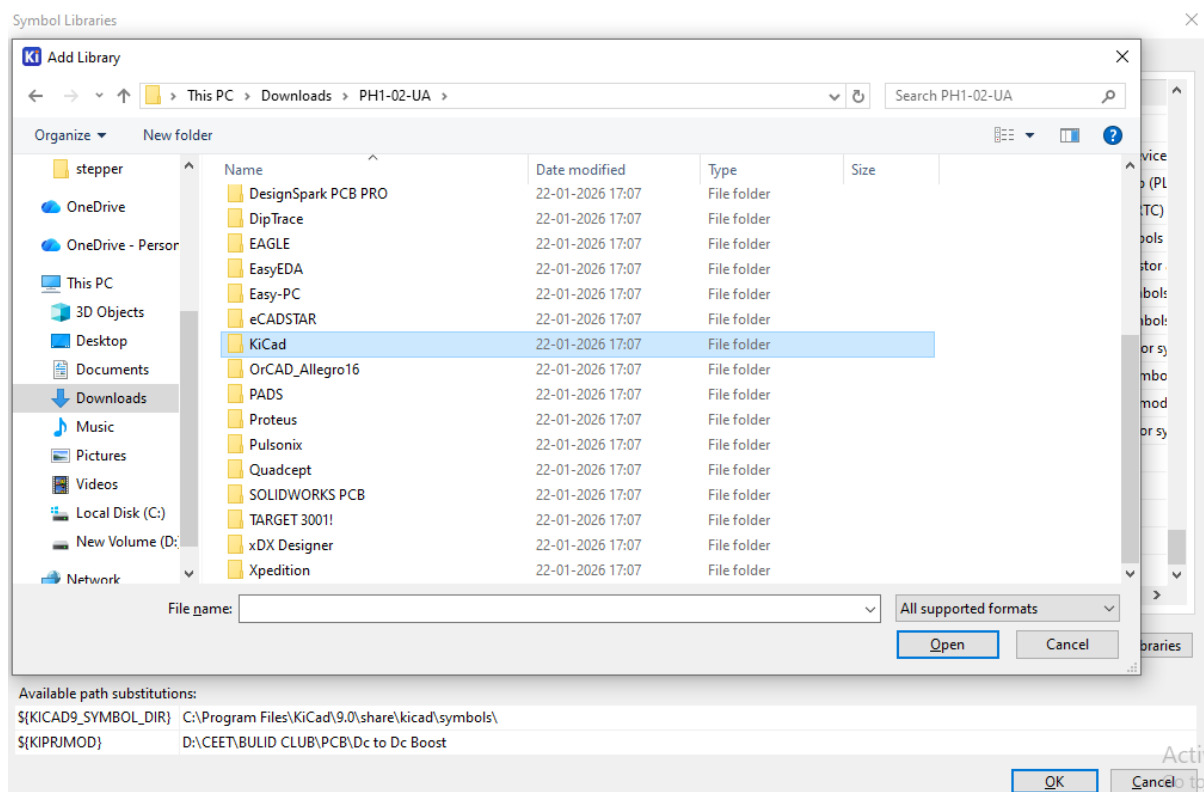
- Add existing library to table option has been open it.



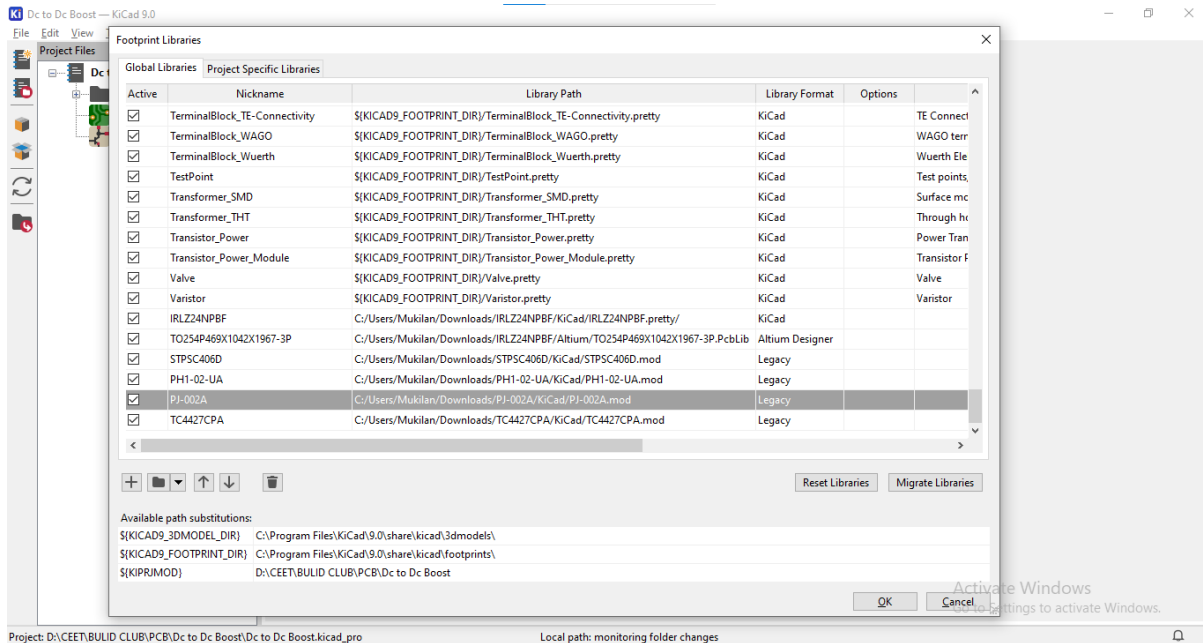
- Select the legacy



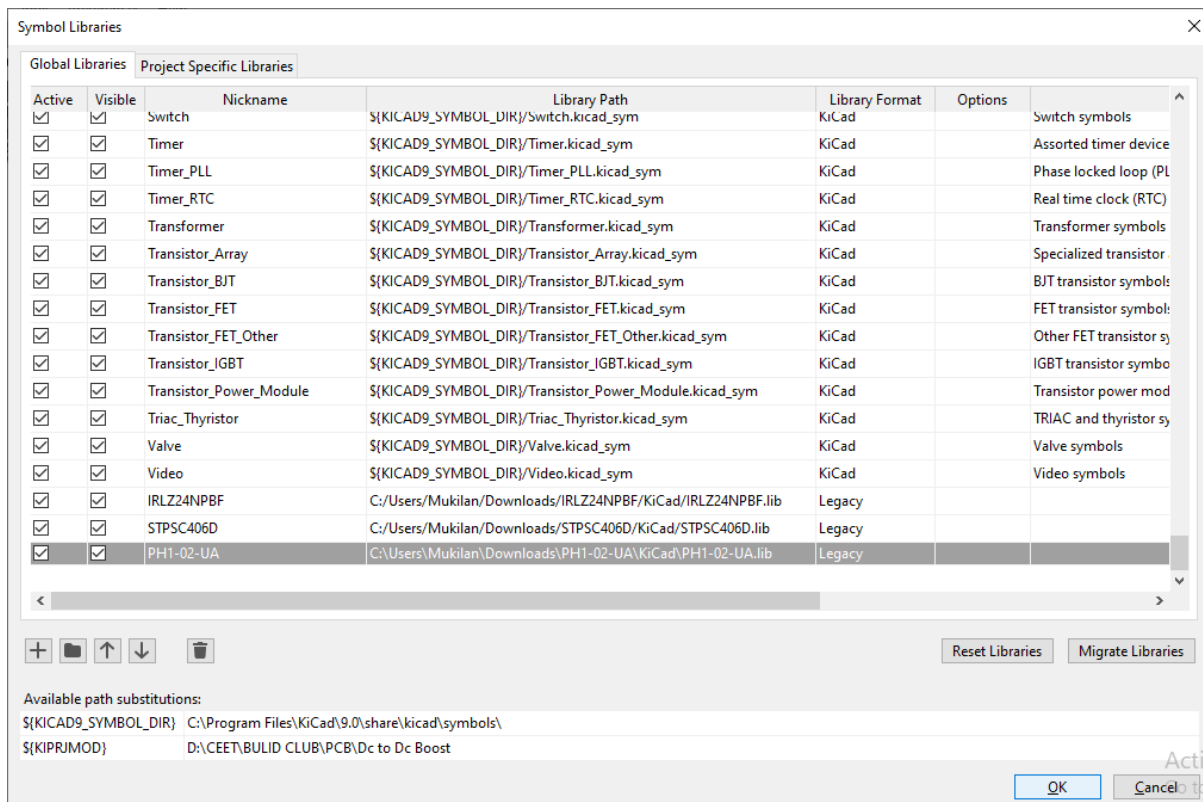
- Open the components folder, select the kicad folder.



- Select the Kicad Library file



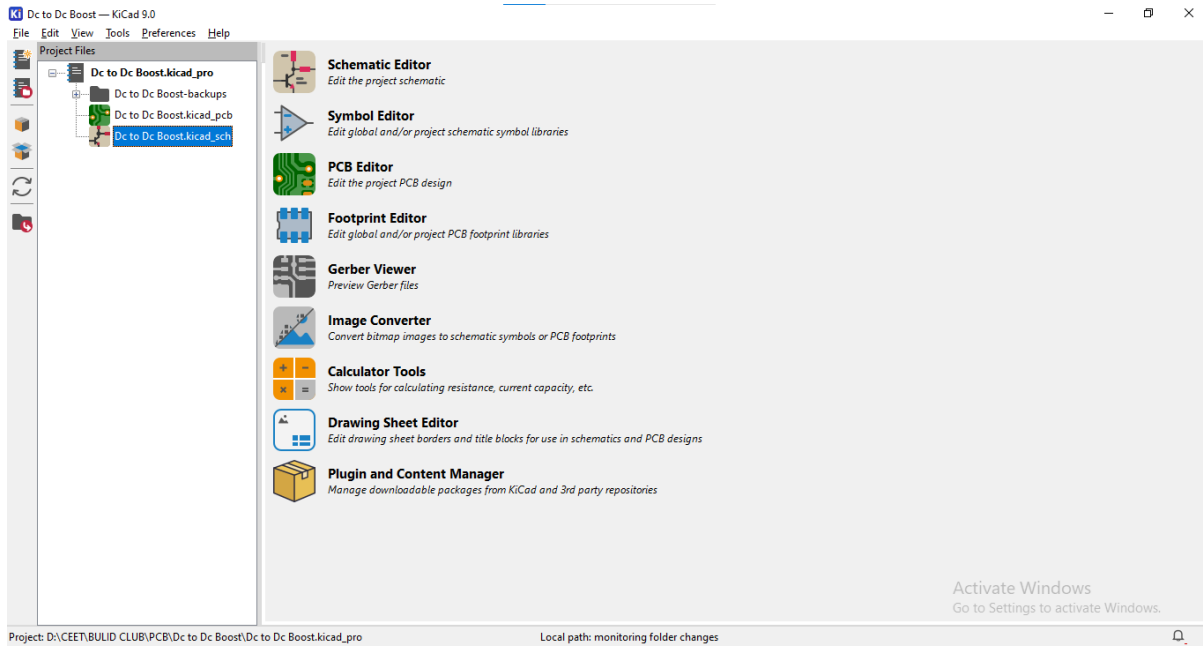
- After Footprint Libraries window has ok option click it.



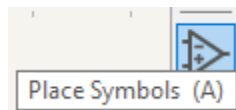
Please follow all the steps to import all components

Step 7:

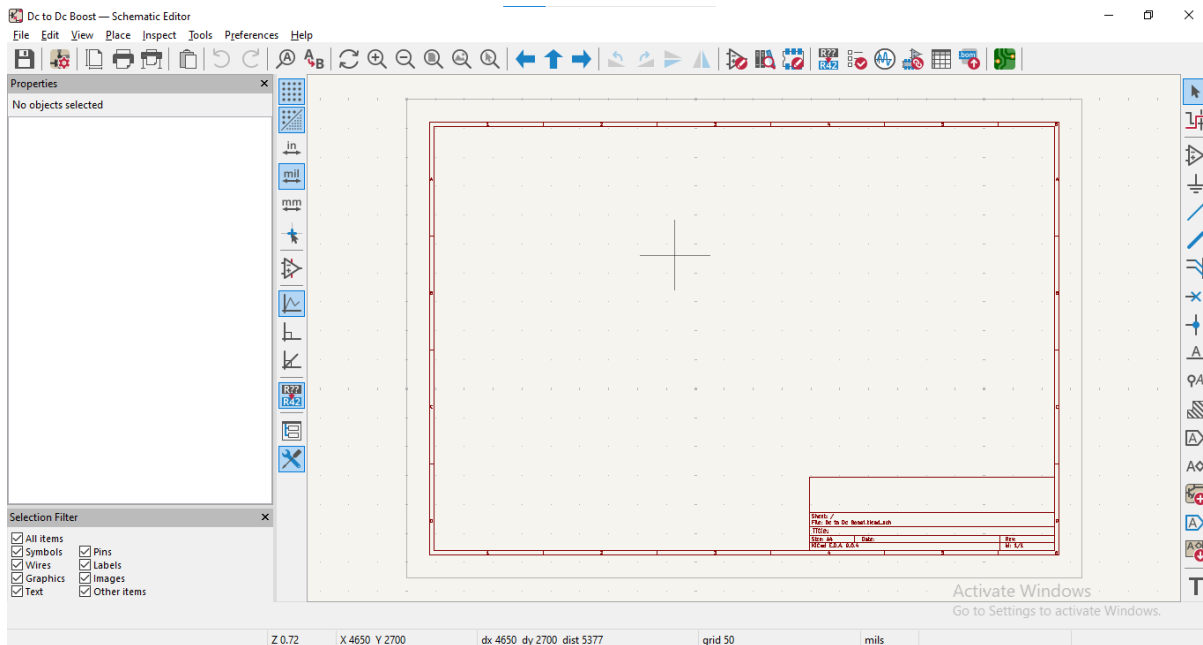
- Open the schematic editor file



- Click the Place Symbols ion,



- On your right-side top corner / press 'A' key word



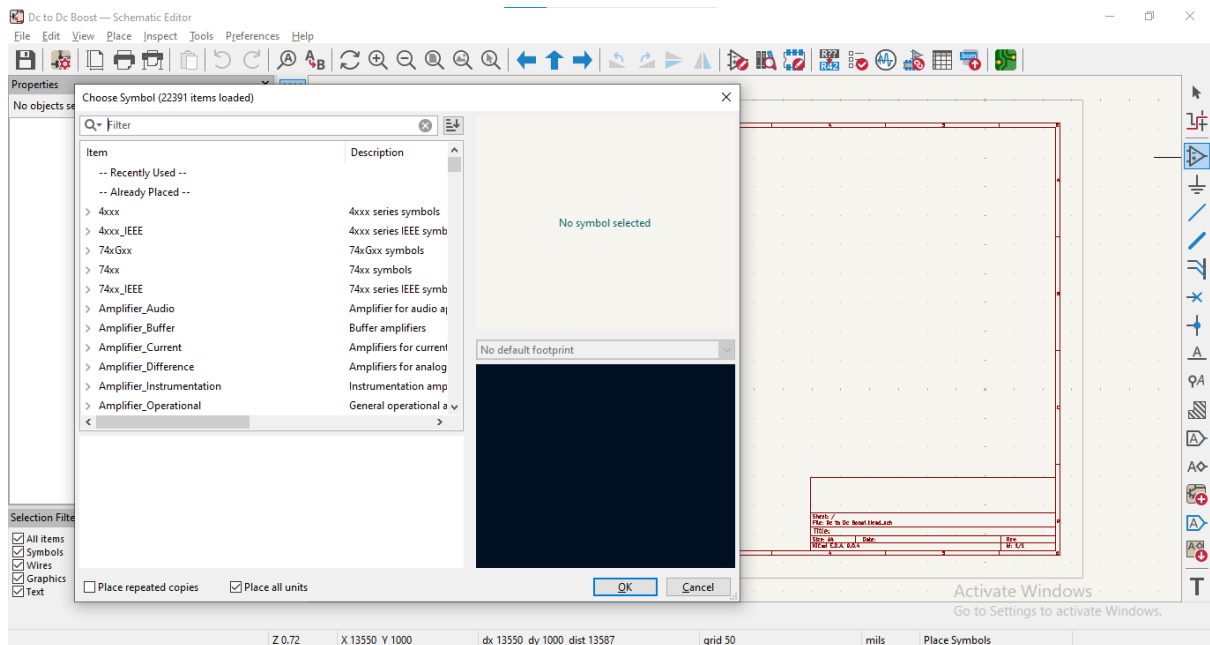
Step 5:

- Select your circuit components

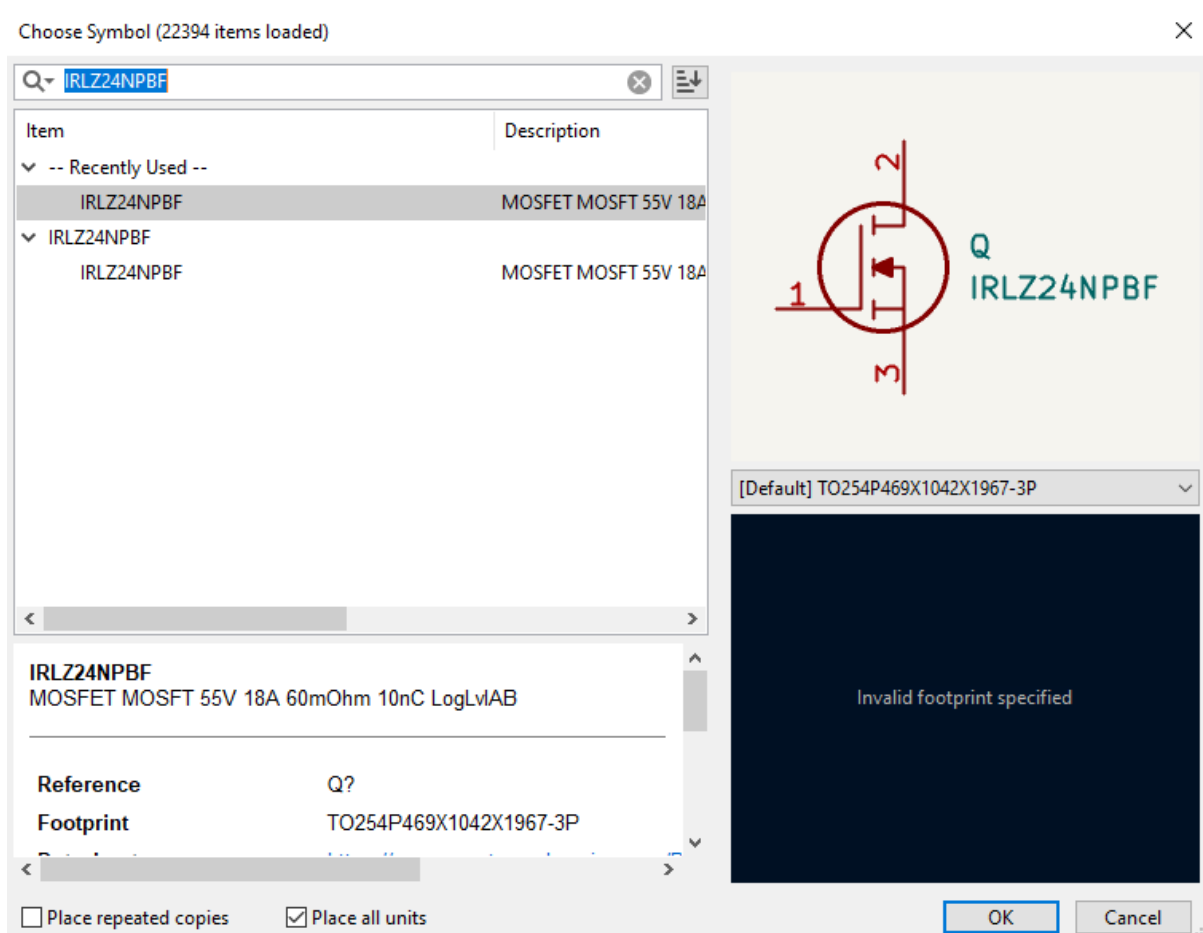
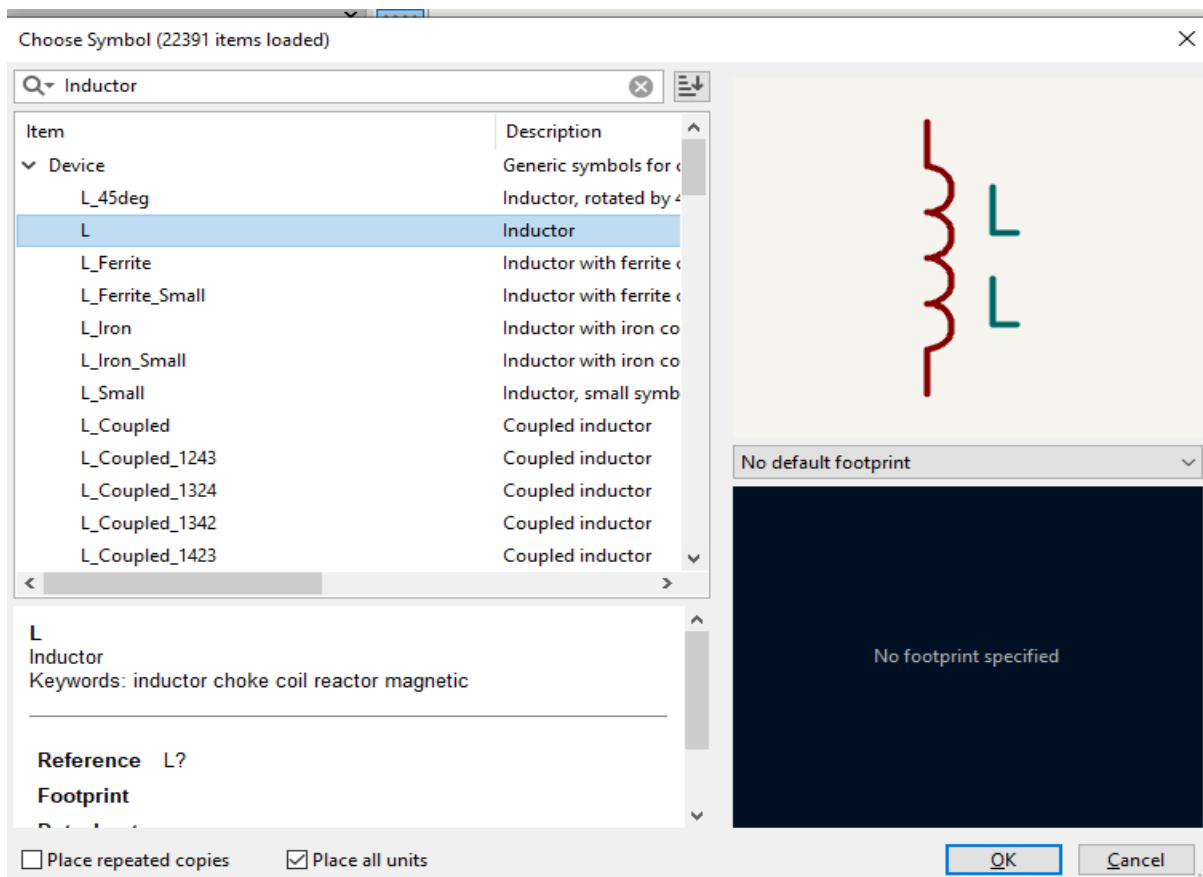
S. No	Components	Quantity
-------	------------	----------

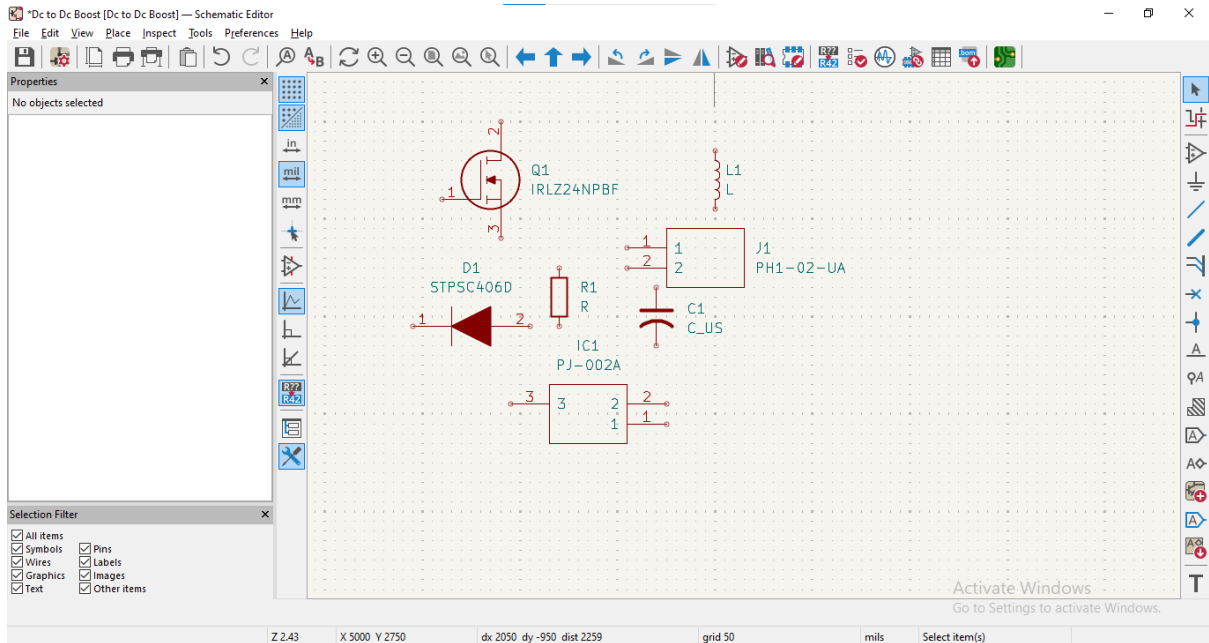
1.	2.2 mH Inductor	1
2.	100-470uf (63V) Capacitor	1
3.	22 ohms,2.2K ohm (Through hole) Resistor	2
4.	IRLZ24NPBF Mosfet	1
5.	STPSC406D Schottky Diode	1
6.	TC4427 Gate Drivers	
7.	PJ-002A Power Connector	1
8.	PH1-02-UA Berg strip male connector	2

- Go to filter option
- Search the components name or part number (ex-inductor or IRLZ24NPBF)

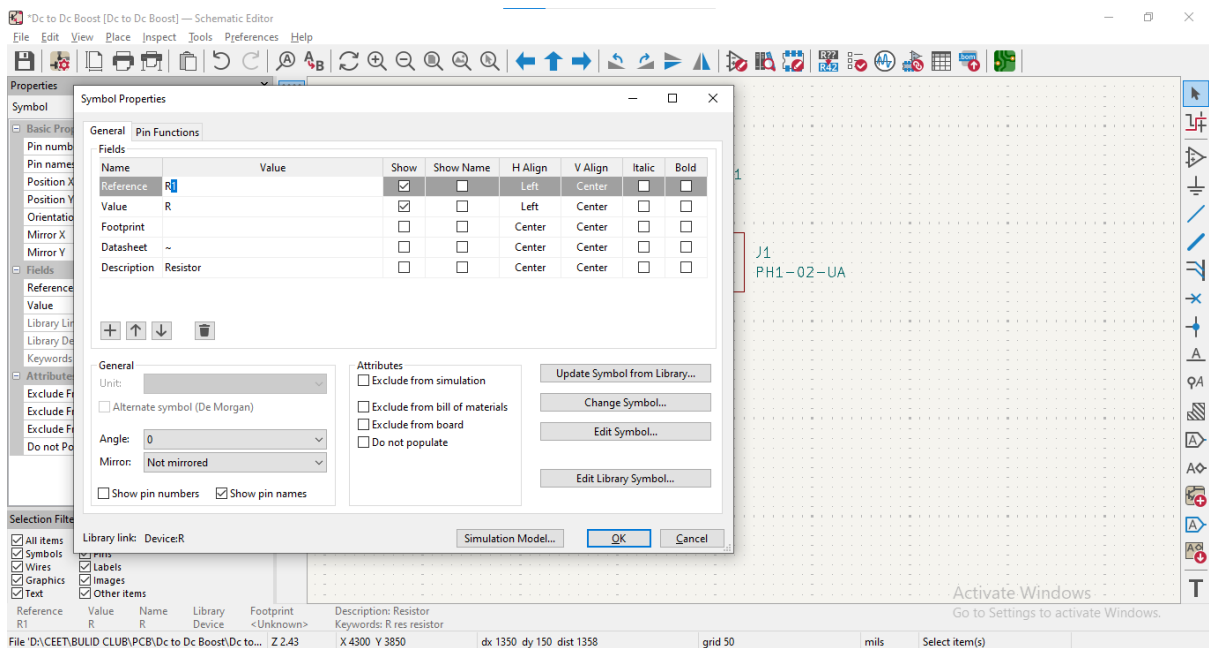


- Select your component after placing it





- To place all components on the sheet
- Select the component press 'E' add the value (Resistor, capacitor and inductor) and all components add the footprint



- Give the value like that (2.2k ohms or 100uf)

*Symbol Properties

General Pin Functions

Fields

Name	Value	Show	Show Name	H Align	V Align	Italic	Bold
Reference	R1	☒	☐	Left	Center	☐	☐
Value	2.2K	☒	☐	Left	Center	☐	☐
Footprint		☐	☐	Center	Center	☐	☐
Datasheet	~	☐	☐	Center	Center	☐	☐
Description	Resistor	☐	☐	Center	Center	☐	☐

+ ↑ ↓ 

General

Unit:

☐ Alternate symbol (De Morgan)

Angle:

Mirror:

☐ Show pin numbers ☒ Show pin names

Attributes

☐ Exclude from simulation

☐ Exclude from bill of materials

☐ Exclude from board

☐ Do not populate

Update Symbol from Library...

Change Symbol...

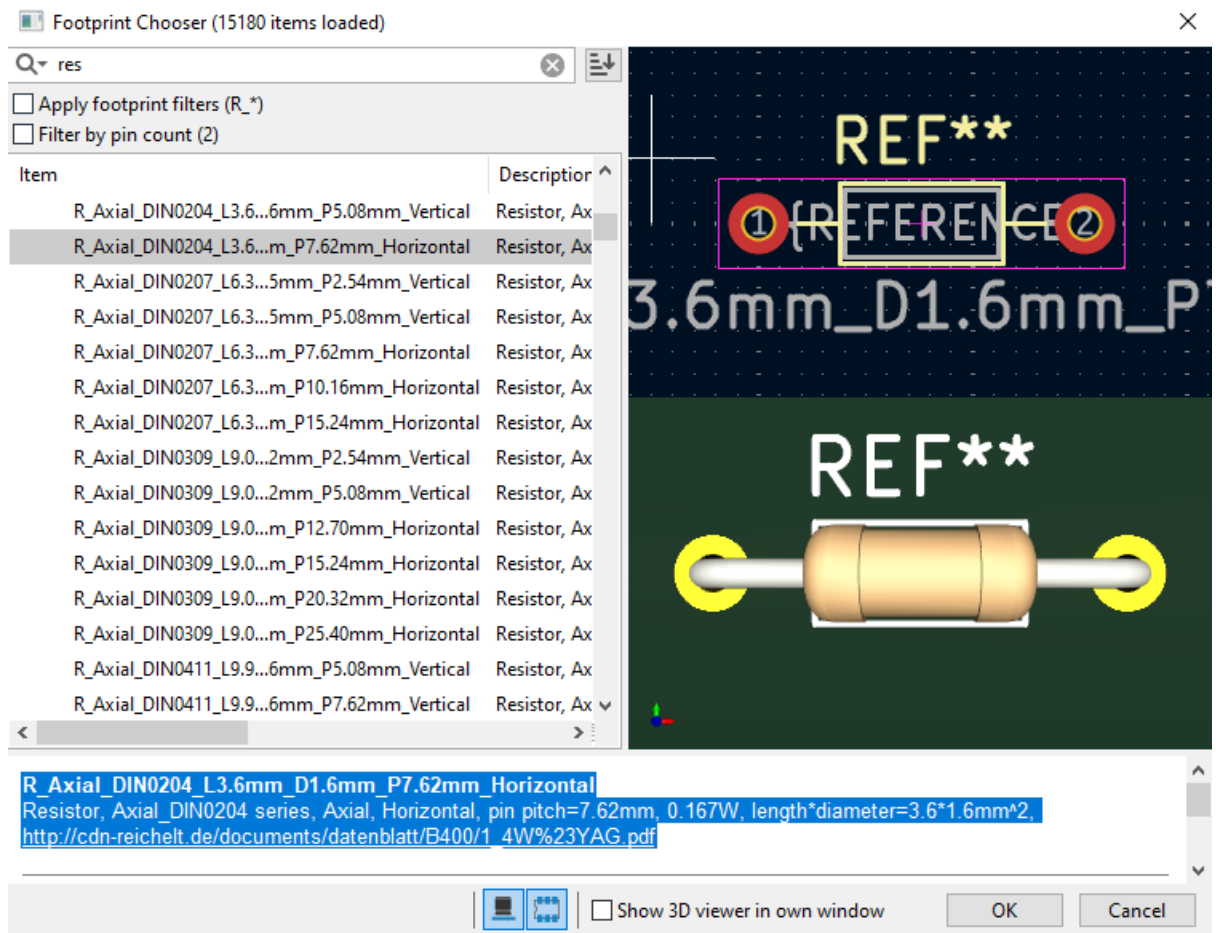
Edit Symbol...

Edit Library Symbol...

Library link: Device:R

Simulation Model... OK Cancel

- Press the footprint option
- Select the part number (like a package-through hole)
- Go to filter option
- Search the components name or part number



- Select your component Footprint after ok it
- Symbol Properties finally ok option click

*Symbol Properties

General Pin Functions

Fields

Name	Value	Show	Show Name	H Align	V Align	Italic	Bold
Reference	R1	☒	☐	Left	Center	☐	☐
Value	2.2K	☒	☐	Left	Center	☐	☐
Footprint	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.1	☐	☐	Center	Center	☐	☐
Datasheet	~	☐	☐	Center	Center	☐	☐
Description	Resistor	☐	☐	Center	Center	☐	☐

+ ↑ ↓ 🗑️

General

Unit:

☐ Alternate symbol (De Morgan)

Angle:

Mirror:

☐ Show pin numbers ☒ Show pin names

Attributes

☐ Exclude from simulation

☐ Exclude from bill of materials

☐ Exclude from board

☐ Do not populate

Update Symbol from Library...

Change Symbol...

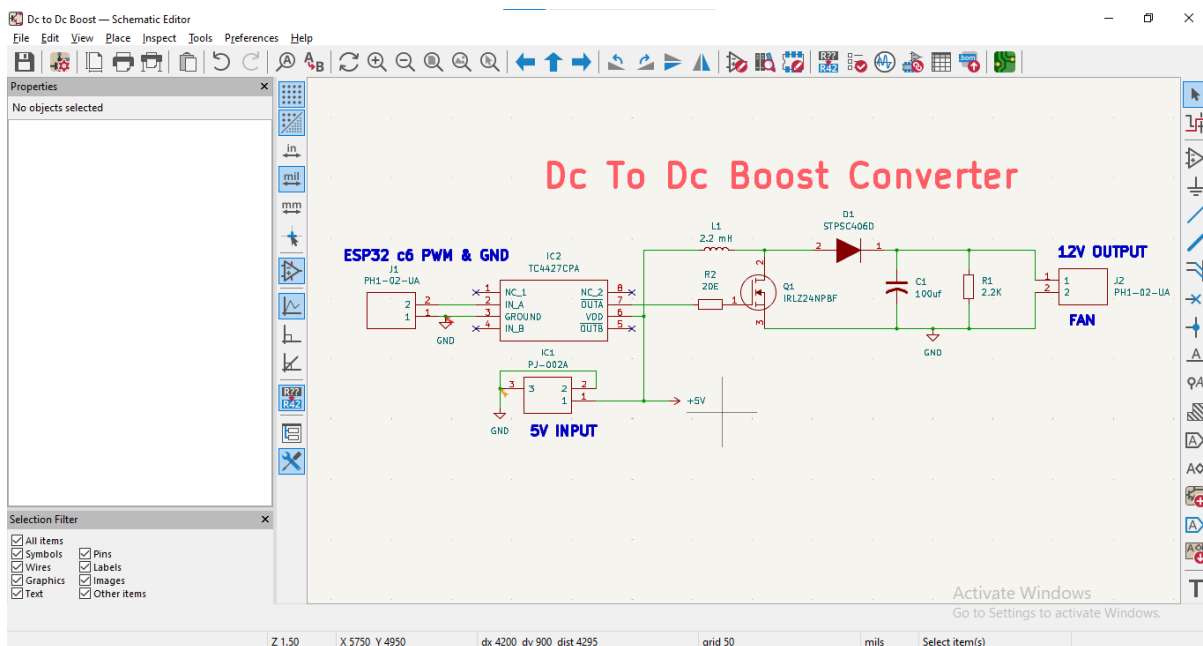
Edit Symbol...

Edit Library Symbol...

Library link: Device:R

Simulation Model... OK Cancel

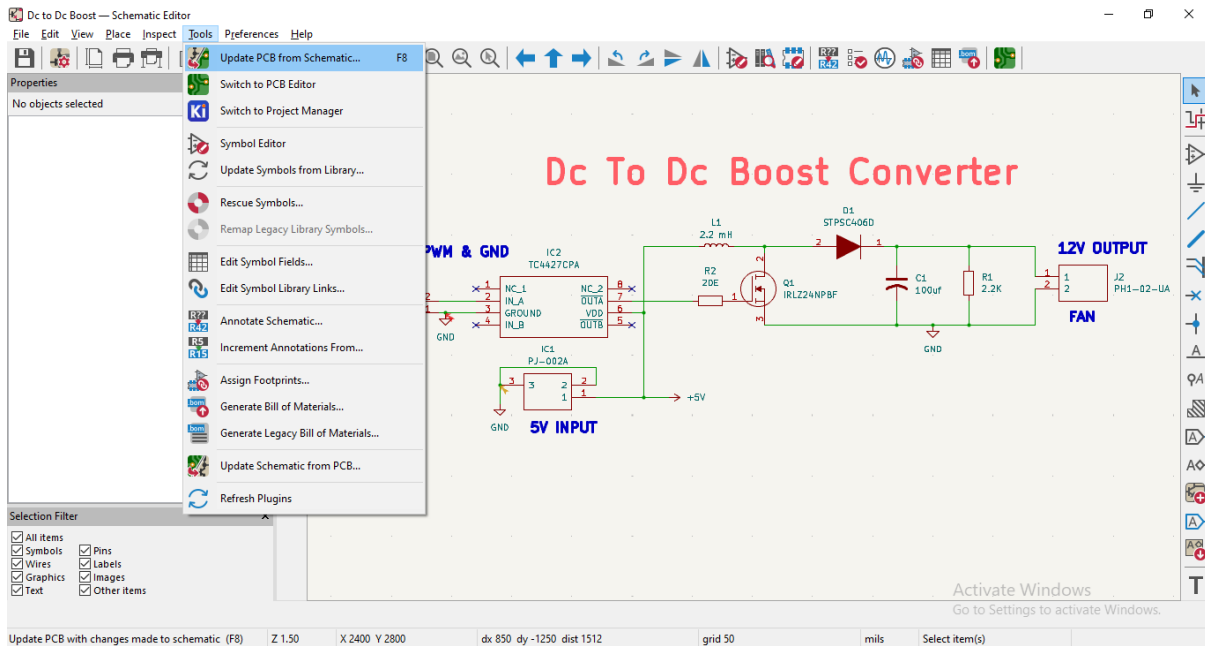
- Next give all components value has enter symbol properties
- Connect symbols with wires
- Press W key word



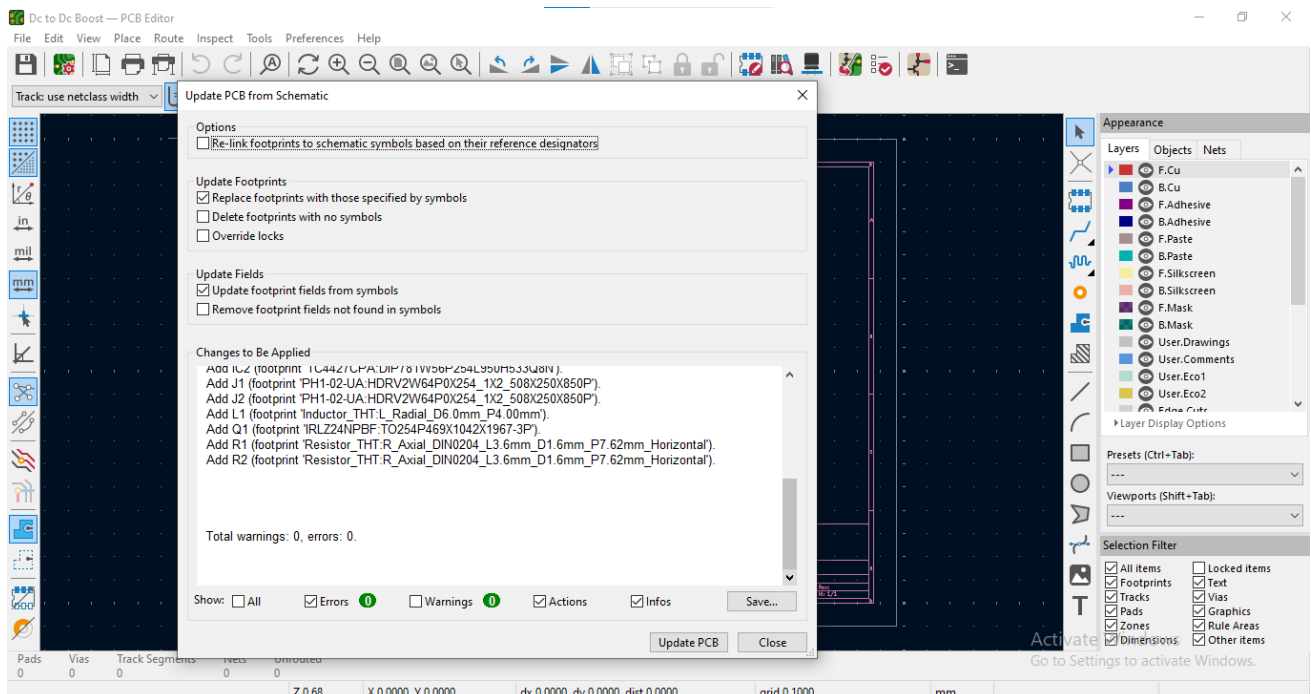
- Save the Schematic file.

Step 8:

- Go to 'Tool' → 'Update PCB from schematic' / F8



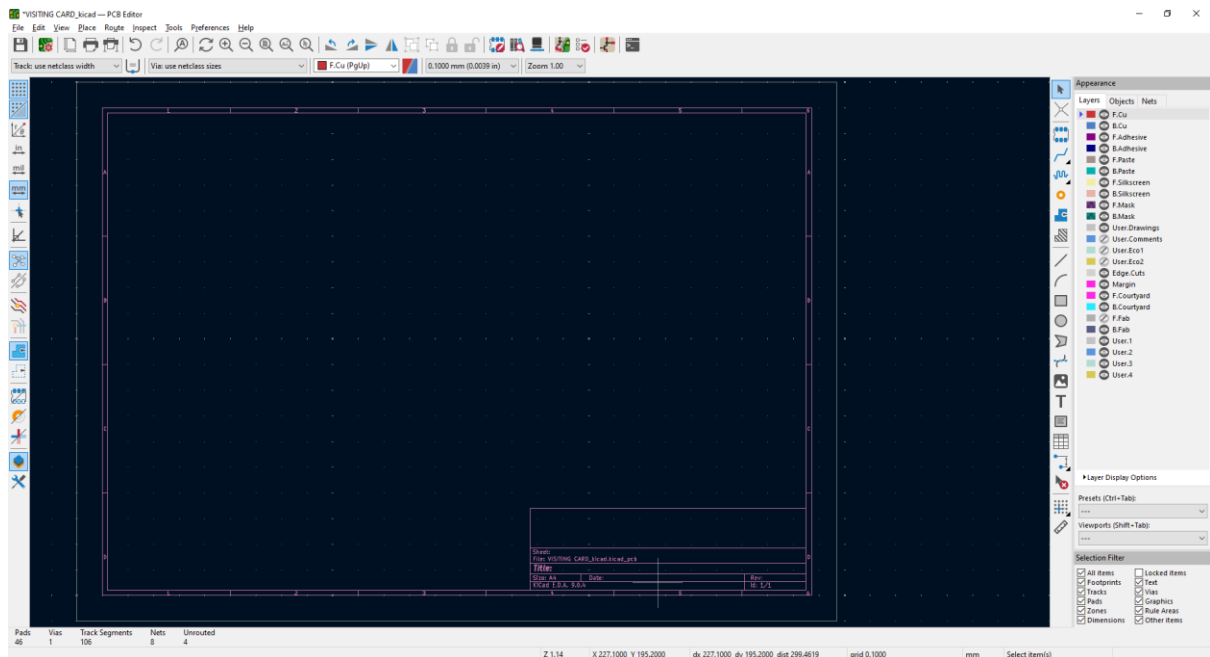
- Check each schematic symbol must have a matching PCB footprint
- If any footprint has missing it can't update PCB, it's shown error warning



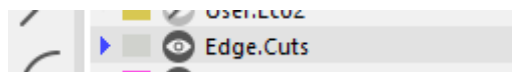
- Go to clear the error (like a No unconnected, annotate symbols, Fix wiring errors, missing power flags) Otherwise update the PCB option click

Step 9:

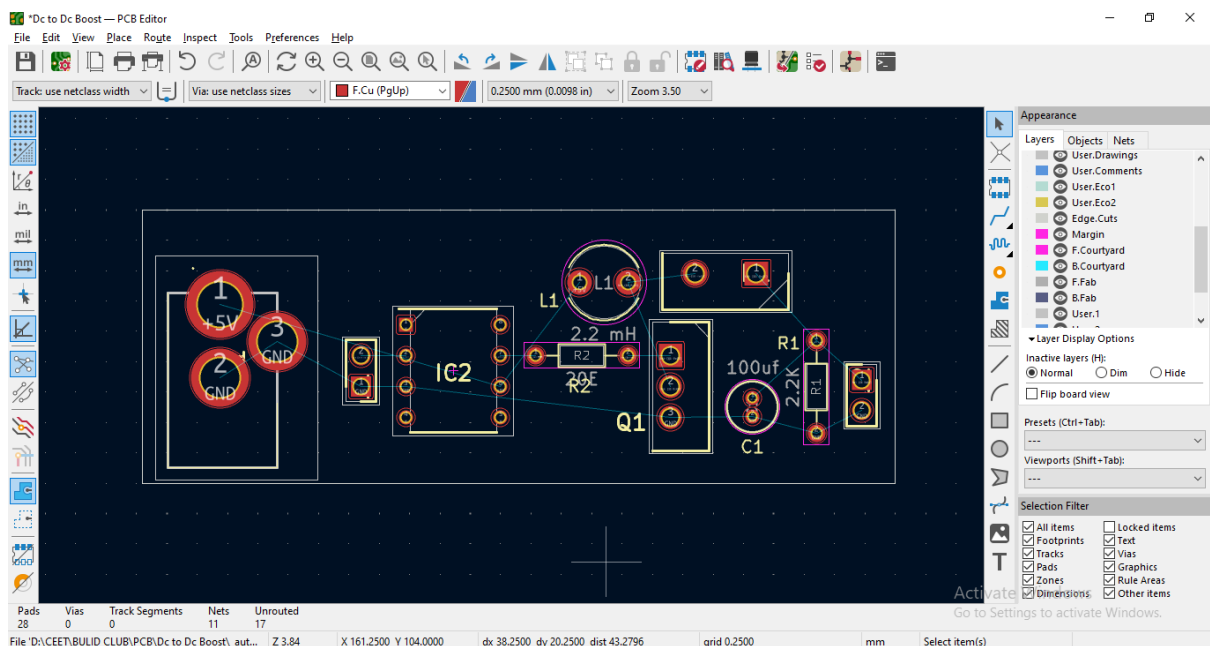
- Set the board outline (Edge Cuts layer)



- Click the Place Symbols icon, On your right-side corner

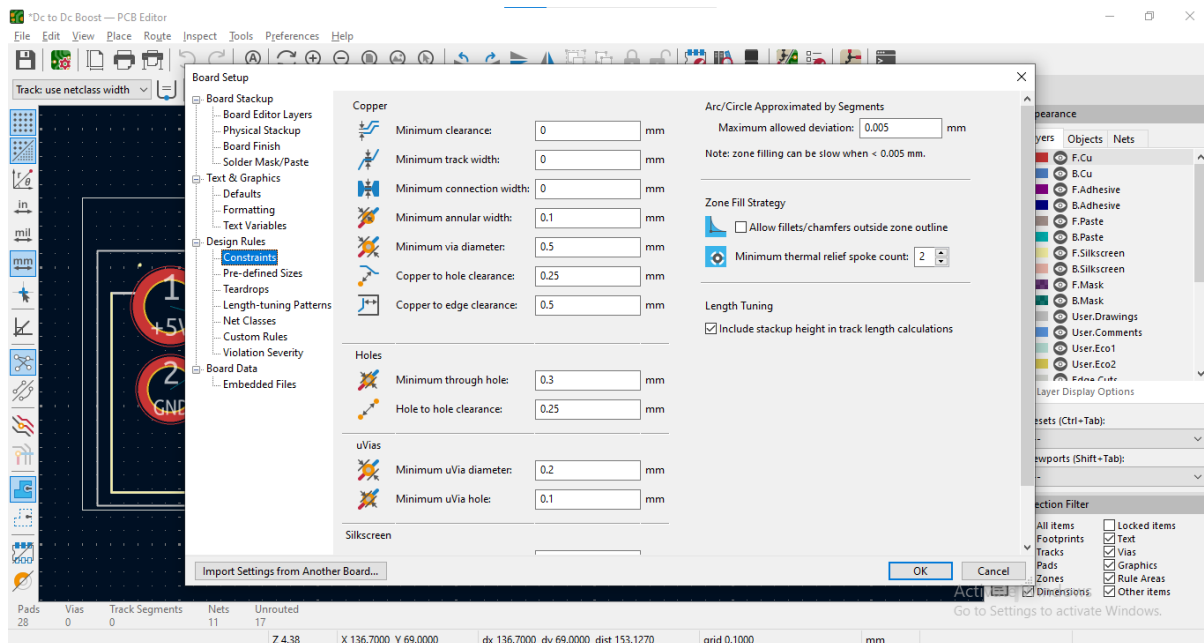
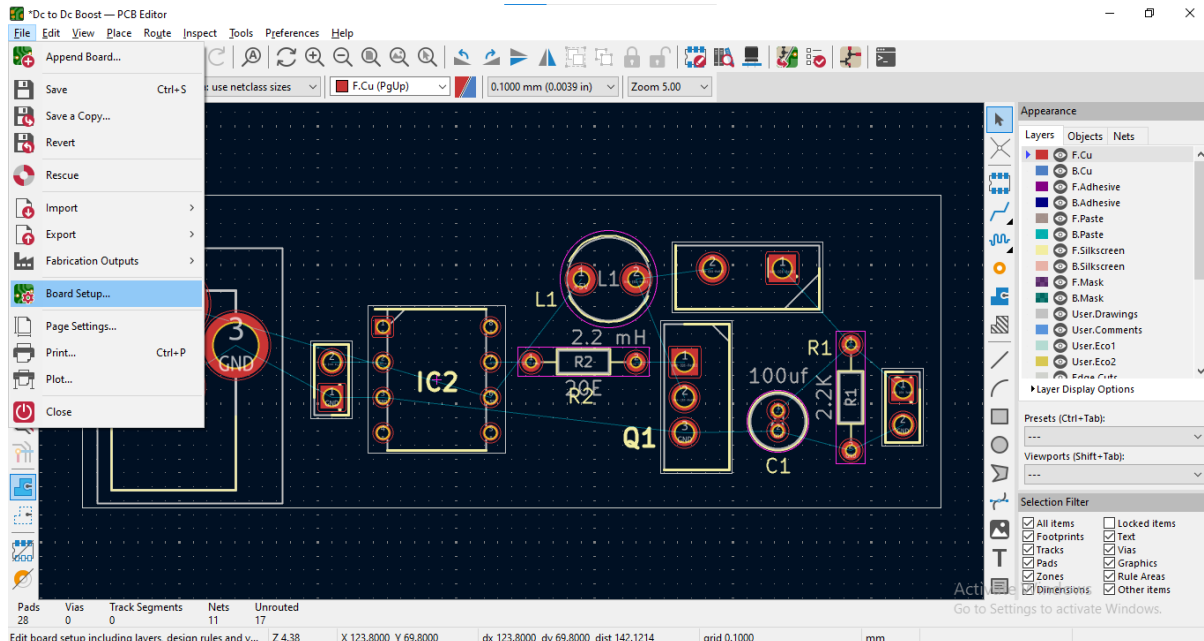


- Select the rectangle tool using to set the board outline
- To place all components on the sheet



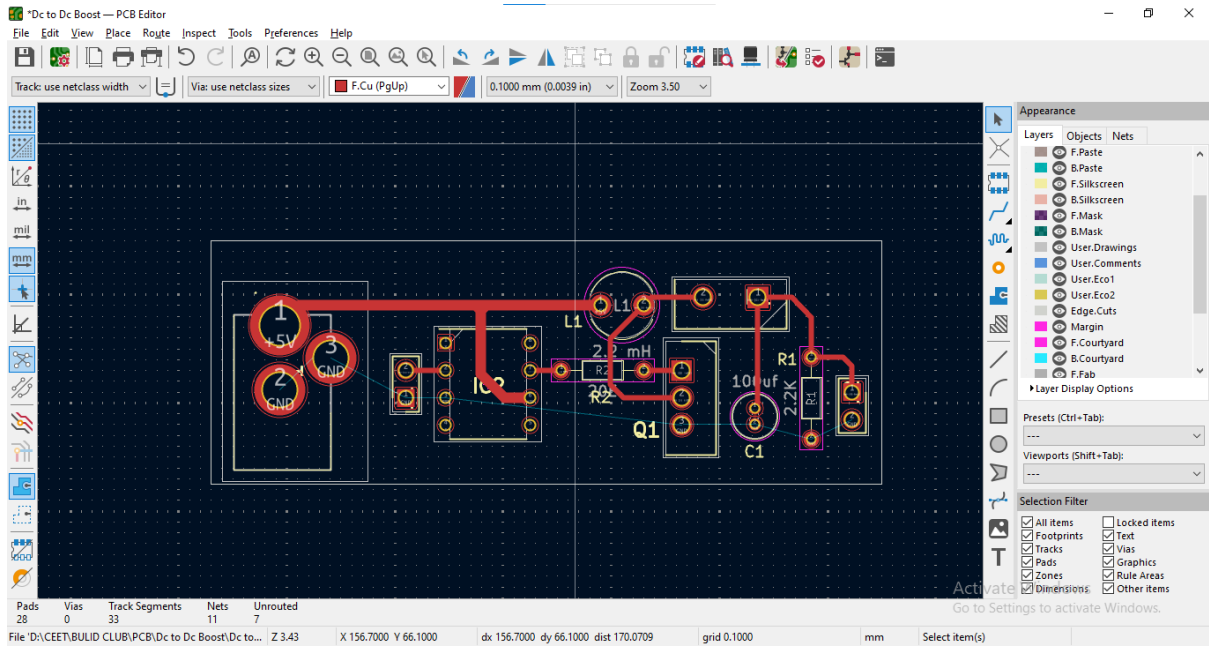
Step 10:

- Next go to File → Board Setup → Constraints
- Set the details of all option
- And refer the [link](#) (**Disclaimer** – ‘Constraints’ depends on PCB manufacturer)



Step 10:

- Go to select the layer after routing in the traces short key “X”
- Traces thicker increase or decrease width via net class selection based

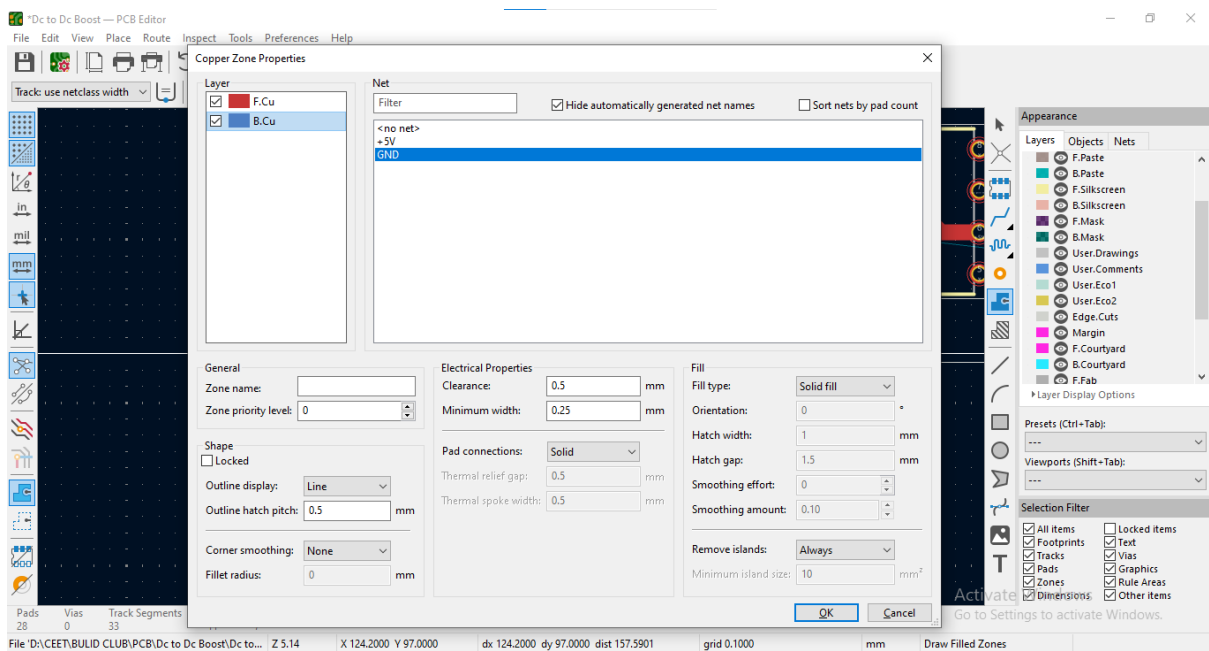


- Click the Place Symbols icon, after click the PCB grid

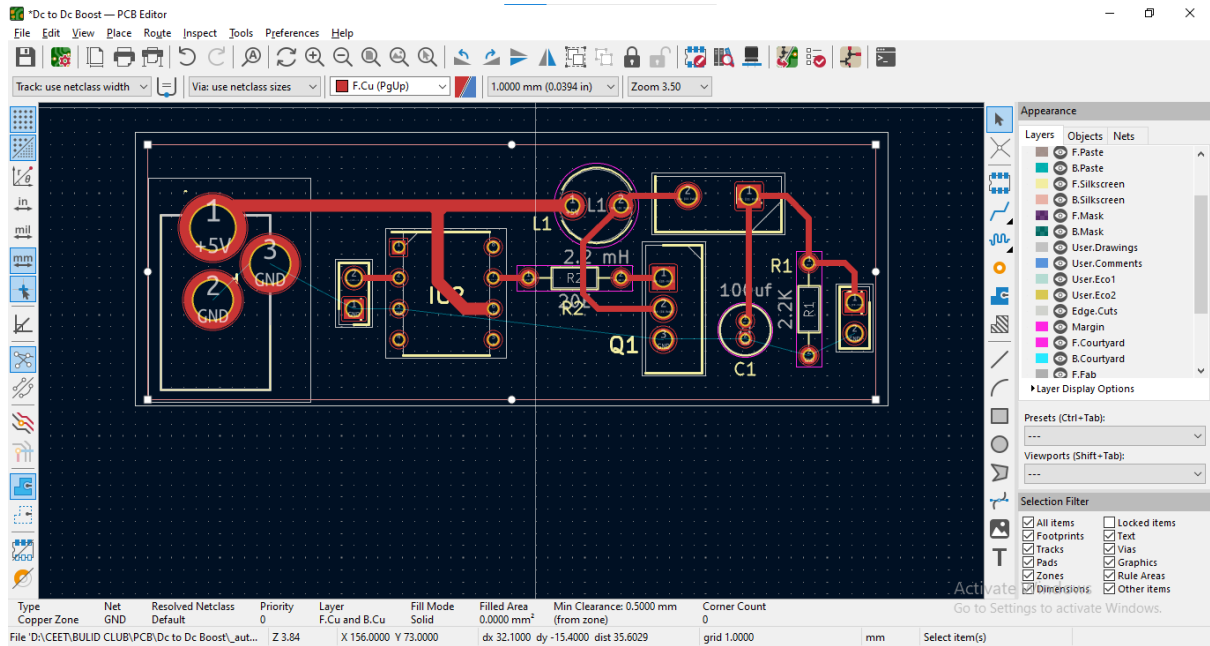


Polygon tool

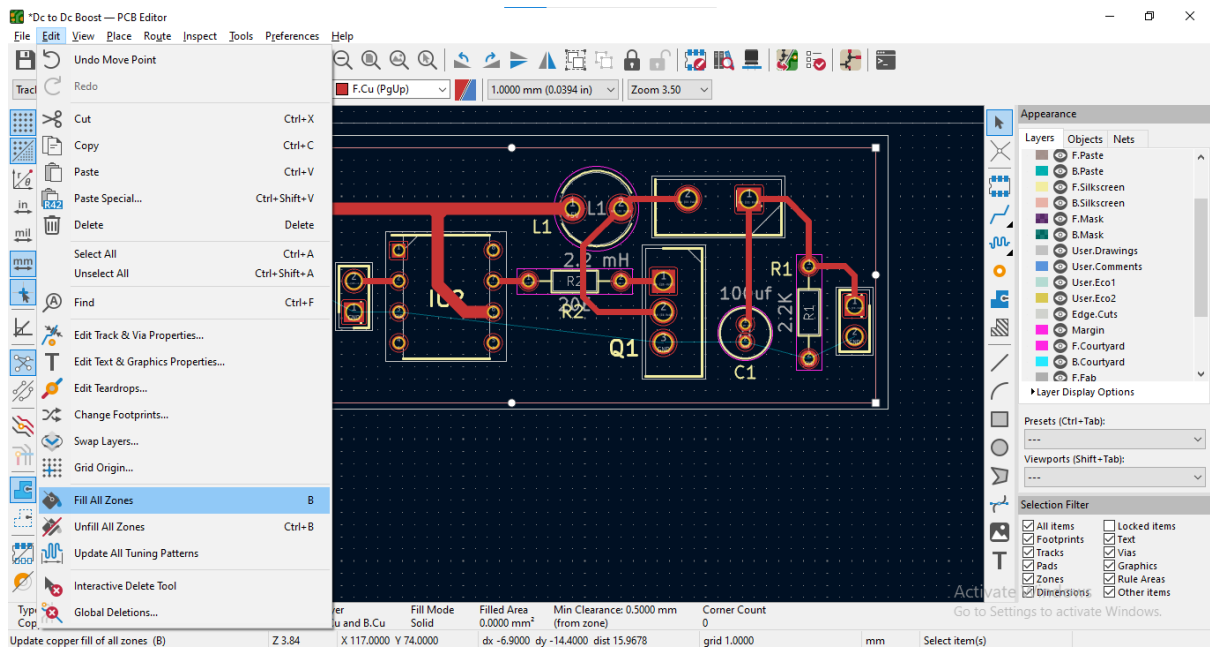
- It's automatically open Copper Zone Properties, selected the GND net

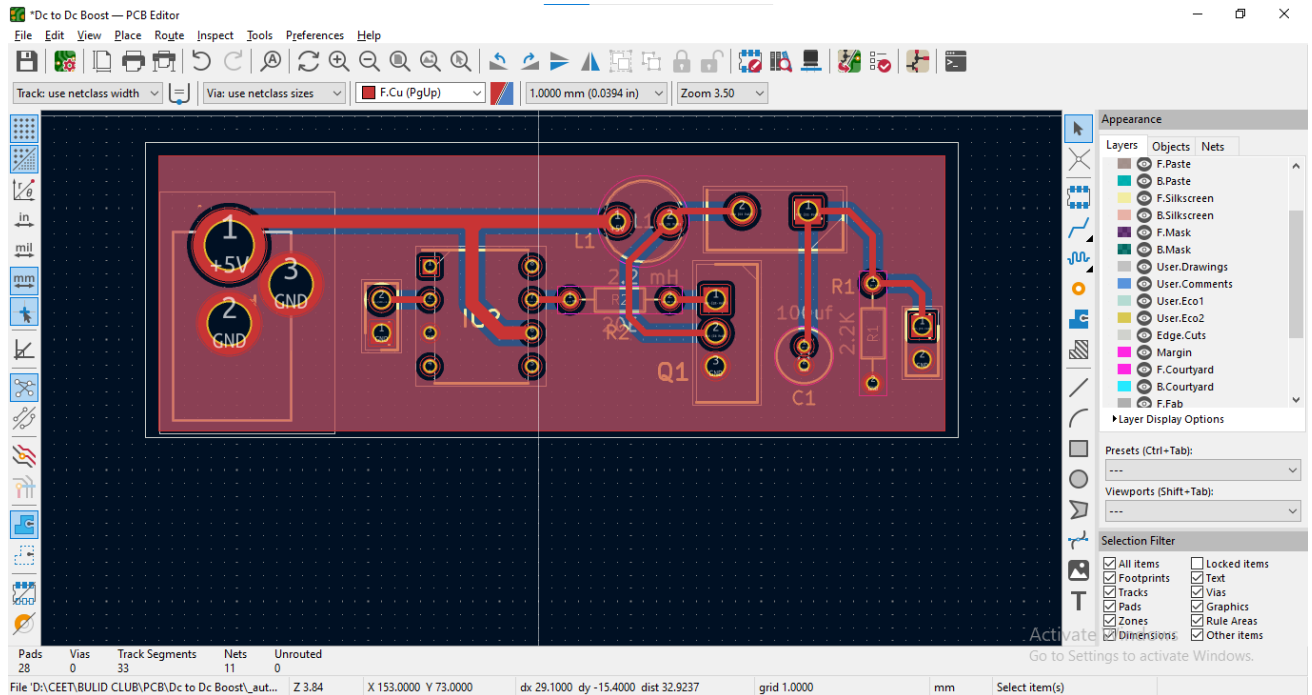


- Polygon using to draw the rectangle



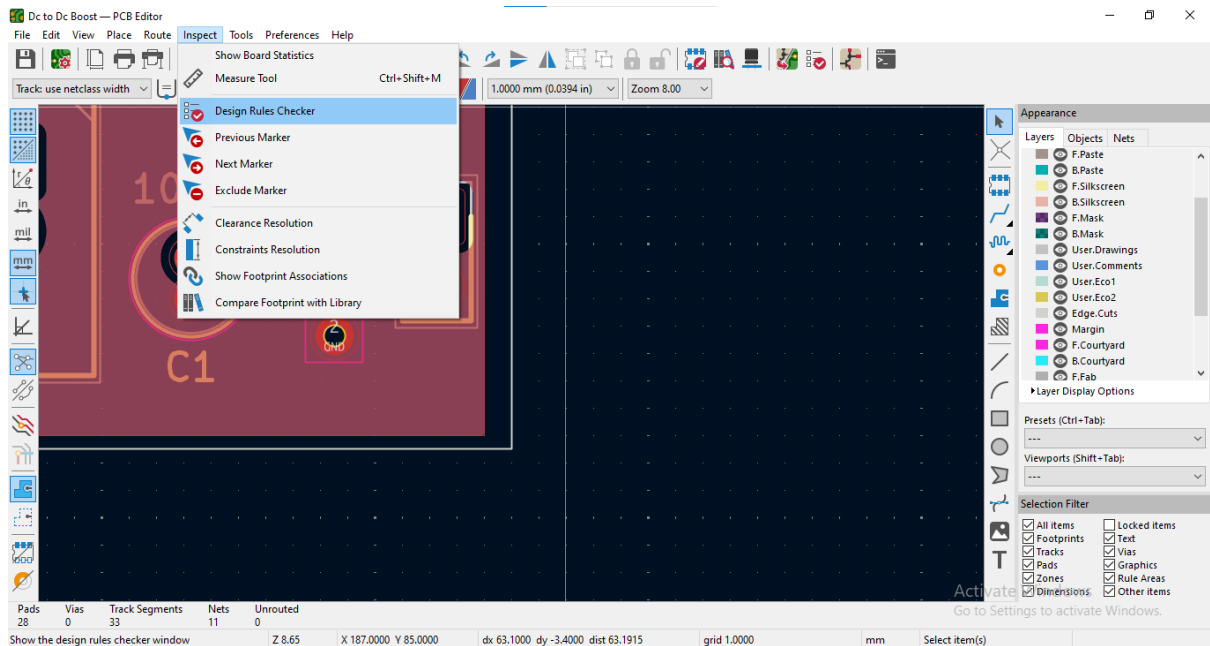
- Press the key “b” or go to edit option → Fill All Zones click it



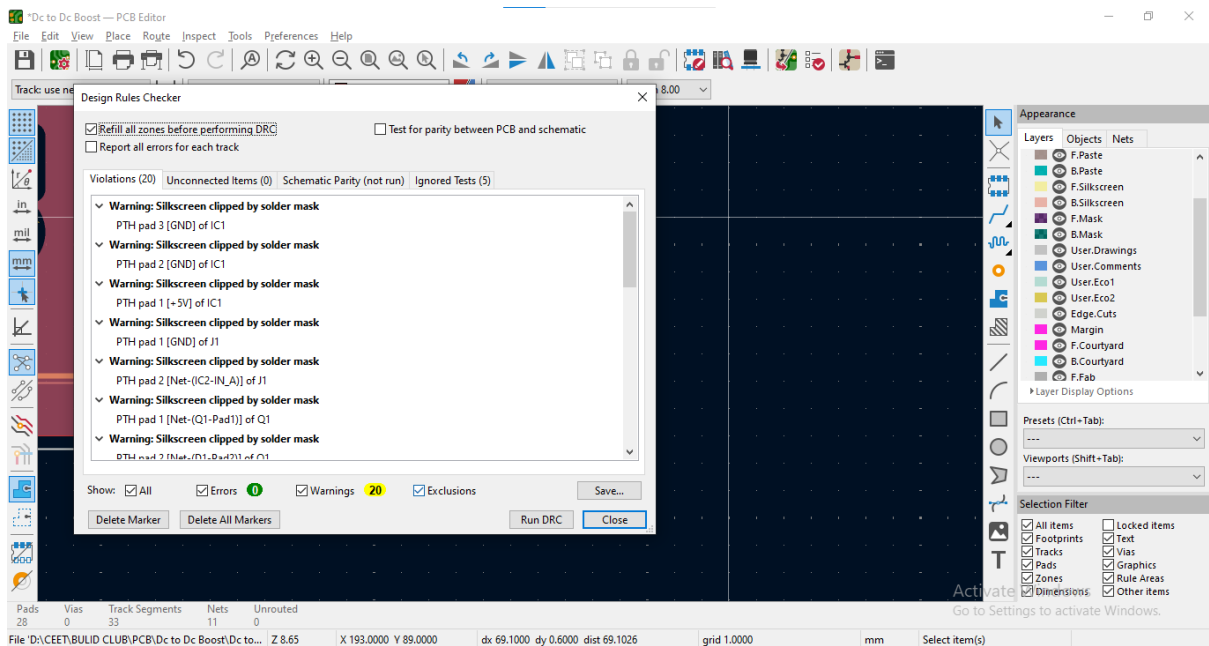


Step 12:

- Go to 'Inspect' → 'Design Rules checker'
- Check each Constraints value must have a matching PCB design
- If any Constraints value has missing it can't run DRC, it's shown error warning Go to clear the error (clearances, unconnected pins, minimum annular rings)

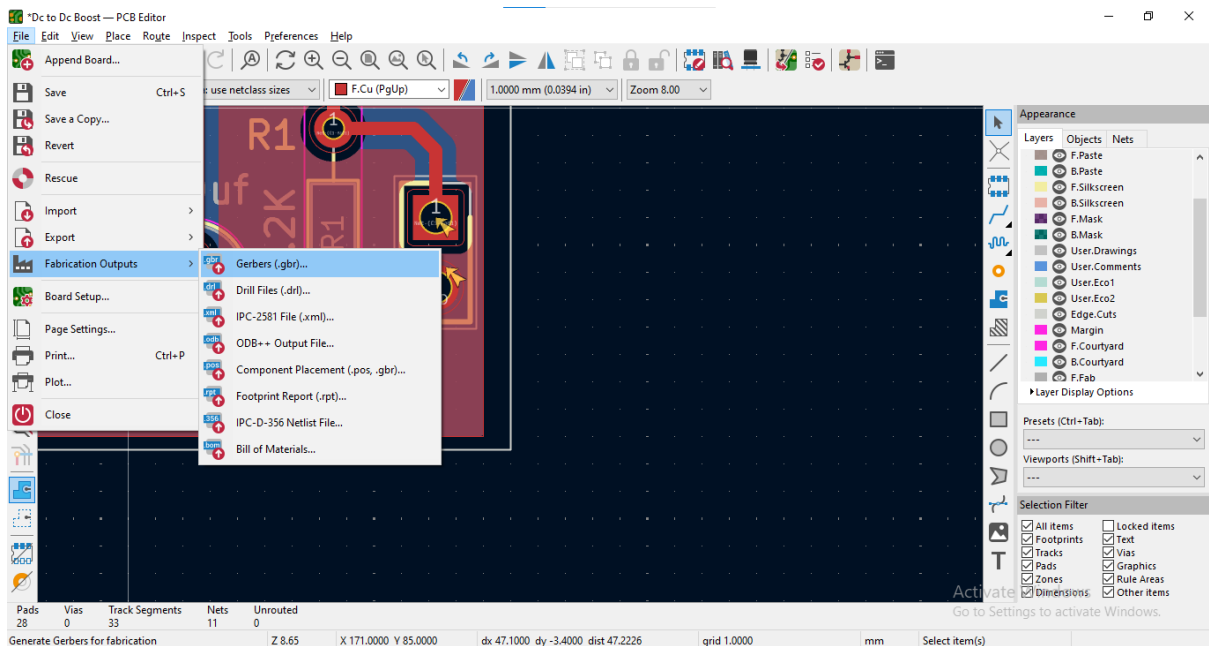


- If clear all error go to Fabrication outputs



Step 13:

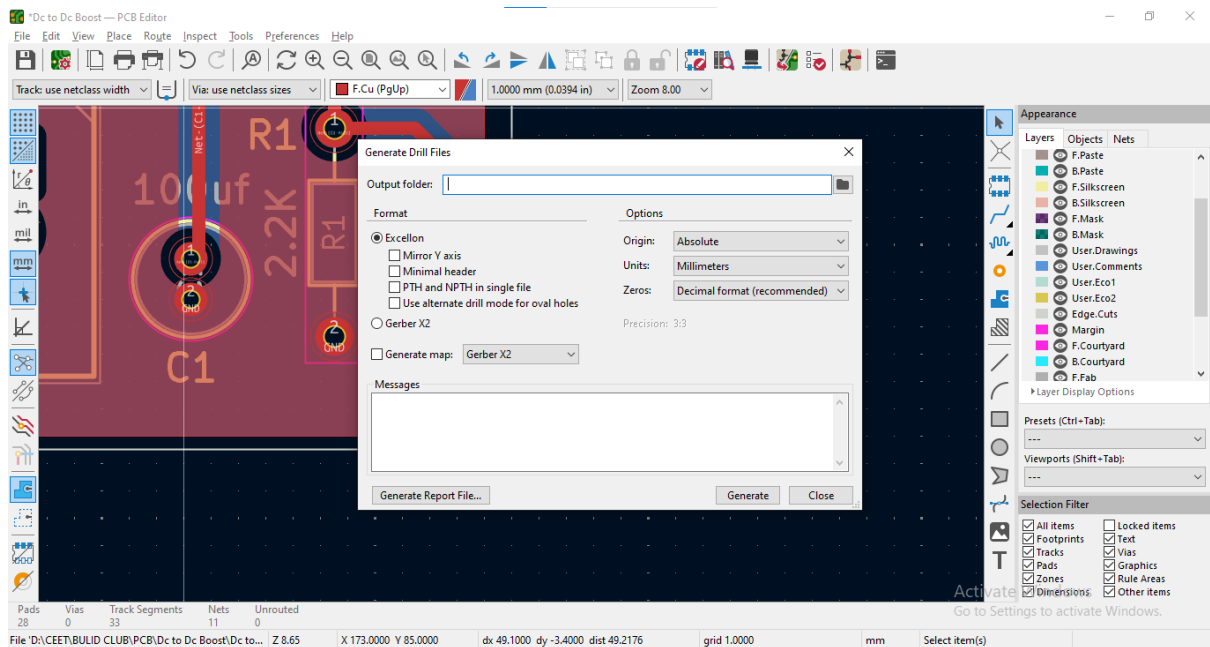
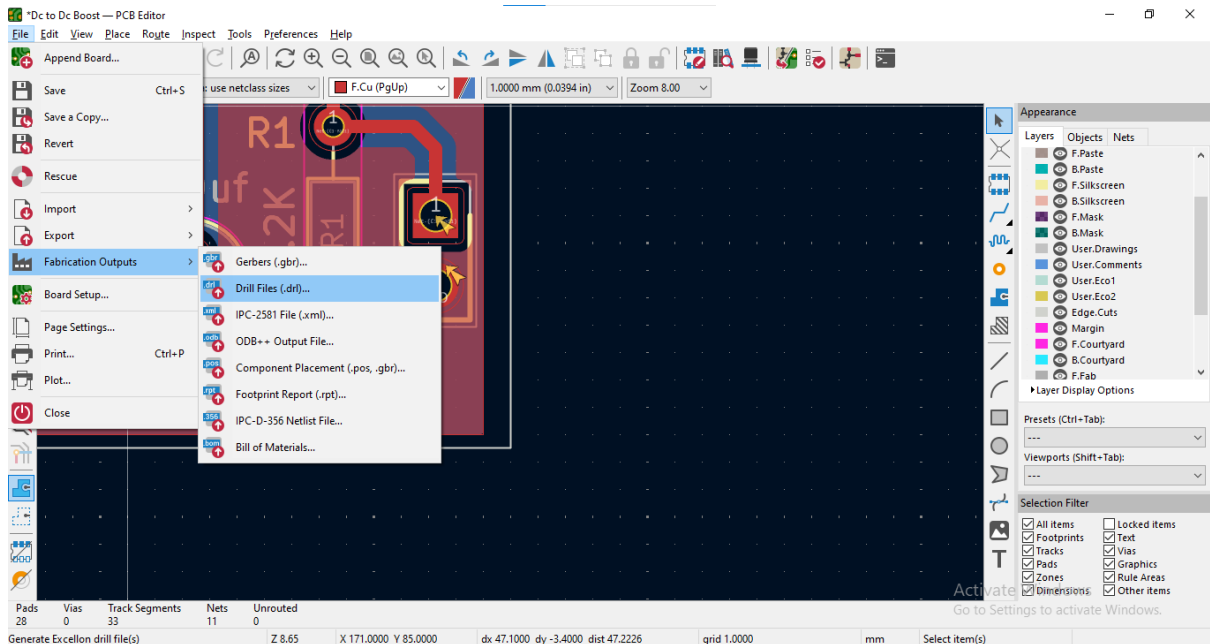
- Go to file → Fabrication outputs → Gerber, click it



- Select all layers
 - F. Cu (Top copper)
 - B. Cu (Bottom copper)
 - F. Silk (Top silkscreen)
 - B. Silk (Bottom silkscreen)
 - F. Mask (Top solder mask)

- B. Mask (Bottom mask)
- Edge. Cuts (Board outline)
- Drill file (Excellon)

- Next Click the Generate Drill Files option



- Finally, Dc to Dc Boost converter PCB is ready
- Please find the reference video link for creating a schematic in KiCad IDE [link](#)
- Reference [link](#)